

The AI Economy: Adoption, Human Validation, and Labor Demand *

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Abstract

I study how human validation requirements shape the response of wages and labor demand to AI adoption. I introduce fixed, recurring validation labor into a general equilibrium task model with endogenous AI adoption. It features a validation threshold determining whether the labor created by AI adoption offsets the productive labor it displaces after accounting for demand expansions. I estimate the model using U.S. occupation-task data on human versus AI involvement at work, and AI adoption rates. The estimated U.S. validation requirement lies below the threshold, implying that productive labor displacement dominates and wages fall. For the occupations in the estimation sample, a ten year projection of declining AI costs raises adoption from 18 to 62 percent and nearly triples output, while reducing employment by 13 percent and wages by 43 percent. Extending the exercise to the entire U.S. economy attenuates the employment and wage declines to 5 and 18 percent, respectively.

Keywords: Artificial Intelligence, Technology Adoption, Validation Labor, Labor Demand, Task-Occupation Labor

JEL Classification: E17, E24, J22, J23, J31, O33

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1 Introduction

Generative artificial intelligence (AI) has rapidly become part of the workplace. Data from the Real Time Population Survey (RPS) shows that the share of U.S. employed adults reporting the use of AI at work increased from 28 percent in 2024Q3 to 45 percent in 2026Q2 [Bick et al. \(2026\)](#).¹ At the same time, governments have explicitly made AI policy a priority. The U.S. AI Action Plan, for example, contains more than 90 federal policy actions encompassing AI innovation, infrastructure development, workforce preparation, national security, and international cooperation. The European Union has also introduced a comprehensive regulatory framework through their AI Act.² In light of these rapid developments, a central economic question remains open: will AI adoption increase labor demand and wages, or will it instead displace workers and reduce wages?

This paper answers this question by starting from a simple premise; the associated workflow of a task, even after AI adoption, may continue to require human labor. Workers may need to inspect the generated output, correct mistakes, verify compliance, authorize decisions, or assume responsibility for the final product. I refer to this recurring input as *human validation labor*. This paper highlights the role of human validation labor that becomes necessary after adoption as a key determinant of the gains or losses from AI at the aggregate level. This paper aims to provide a theoretical framework to study human validation labor and further quantify its role in the response of wages and labor demand.

The contribution of the paper is both theoretical and quantitative. (1) On the theory front, I introduce a fixed and recurring human validation requirement, measured in labor hours, into an endogenous task adoption model. I show that two economies can have the same AI adoption rates, adopt AI for the same tasks and experience similar AI productivity gains, yet exhibit opposite wage and employment responses because their AI workflows require different amounts of human validation. I characterize a minimum validation labor threshold above which AI adoption increases equilibrium wages and employment. (2) Quantitatively, I estimate this threshold for the U.S. and locate its economy relative to it. To do so, I construct an occupation task dataset that links the relevant measures of human and AI involvement by occupation and task, realized AI adoption, labor market outcomes, and task exposure to AI. I use this data, in particular the observed adoption rates to infer the otherwise unobserved cost of operating AI and to structurally estimate the relevant scale parameter governing the level of the estimated human validation hours. The

¹Source: RPS series link [here](#).

²See source articles [here](#) and [here](#).

estimates position the U.S. economy below the threshold, implying that further AI adoption would reduce labor demand and equilibrium wages despite increasing output.

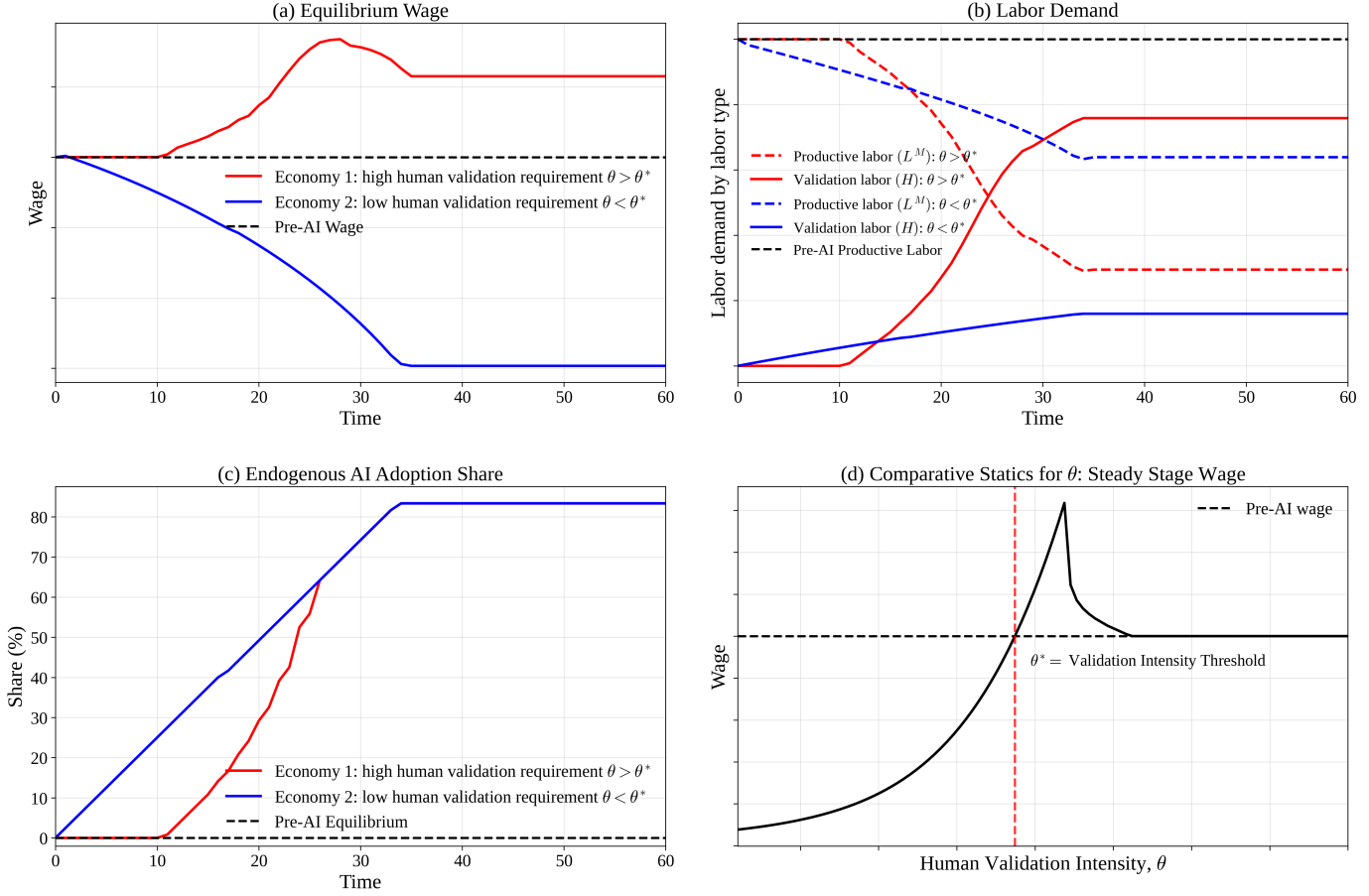
The model is a general equilibrium task model in which each task can be operated manually or with AI. Under the manual method, labor is the productive input, and the amount of labor required increases proportionally with task output. Under the AI method, output is produced using AI compute, but production can take place only after the task specific human validation requirement has been satisfied. This requirement is fixed and recurring: once the required validation has been performed, producing additional units would require more compute but no additional validation labor. Installing the AI method also requires a one time setup labor cost. Firms choose to adopt AI for a task only when the capitalized operating savings exceed this setup cost, and adoption is irreversible. Importantly, adoption, output, the allocation of production across tasks, labor demand, and the equilibrium wage are jointly determined.

An increase in AI adoption, induced by lower AI unit costs, affects productive labor through two opposing forces. First, it eliminates productive labor from the tasks that switch from manual production to AI. Second, the human validation channel introduces labor demand as every AI adopted task creates a recurring demand for review labor. In addition, the resulting decline in production costs lowers the price of the consumption good, expands demand, and increases production of the complementary tasks that remain manual; a scale effect. These effects, validation requirement and scale, may completely offset the displacement of labor. Thus, the wage response ultimately depends on whether the validation labor created by AI is sufficient to replace the productive labor displaced after accounting for the expansion in demand.

This comparison delivers the paper's main theoretical result. As AI displaces productive labor in the aggregate, there exists a human validation threshold below which the validation labor is insufficient to replace the productive hours displaced by AI, causing labor demand, equilibrium employment, and wages to fall. Above it, the creation of validation labor dominates productive labor displacement, causing all three to rise. A third outcome is also possible. If lower prices expand demand for the remaining complementary manual tasks sufficiently, productive labor demand may increase even before validation labor is considered. In that case, AI raises wages and employment regardless of the validation requirement.

Figure 1 illustrates these channels. Panels (a), (b) and (c) compare two otherwise identical economies that face the same decline in AI costs and differ only in their human validation requirements. In the high validation economy, the labor created by validation exceeds productive labor displacement and the equilibrium wage rises. In the low validation economy, productive labor displacement dominates and the wage falls. Nevertheless, both economies eventually reach

Figure 1: AI adoption, human validation labor, and the equilibrium wage



Notes: Panels (a) to (c) compare two economies facing the same staggered decline in AI unit costs. The economies differ only in human validation intensity. Panel (d) reports the steady state wage across alternative validation intensities θ . The figure presents illustrative trajectories. See Figure 2 for actual model simulations.

the same high adoption rate. Panel (d) summarizes this result by showing how the steady state wage changes as an economy crosses its human validation threshold.

To bring the model to the data, I combine five principal data sources characterizing the U.S. AI economy. (1) WORKBank surveys, (2) O*Net activity classifications, (3) RPS Surveys, (4) Current Population Survey (CPS), and (5) AI task exposure data.³ WORKBank provides expert assessments of the human and AI involvement required across tasks and occupations. O*NET supplies the common activity classification and information about the composition of work within occupations. RPS provides realized AI adoption rates by work activity and occupation. The Current Population Survey and Occupational Employment and Wage Statistics provide employment,

³See the data Section 3 for the details.

working hours, and wages. Finally, [Eloundou et al. \(2024\)](#) provide the task AI exposure classifications. The combined data set manages to preserve differences across tasks within an occupation. This is important because workers in the same occupation may use AI for some activities but not others, and those activities may require very different amounts of human validation.⁴ An important data limitation is that the WORKBank sample of occupations represents 15.25 percent of U.S. salaried employment, which [Shao et al. \(2025\)](#) describe as a broad and representative cross section at the sector level.

The estimation proceeds in two steps. First, I use the WORKBank assessments and the O*NET activity structure to estimate relative human validation requirements across occupations and activities. Second, I jointly structurally estimate the common scale parameter that converts these assessments into labor hours and the AI unit cost process that rationalizes observed adoption over time by activity and occupation. In other words the model uses adoption rates as targets to discipline and reveal the economic cost of AI that firms face when deciding whether to automate a task. The estimated validation requirement is approximately 20.7 hours per adopted occupation-task pair per quarter. The corresponding validation threshold exceeds 400 hours, thus placing the U.S. AI adopting sector of the economy on the labor displacement side of the threshold: human validation creates labor demand, but not enough to replace the productive labor displaced by AI.

The estimated AI unit cost declines at approximately 22.7 percent per year. Using this fact I extrapolate the decline of AI unit costs until 2035 to construct a conditional ten year forecast of the U.S. AI economy. The conditional forecast predicts an increase of AI adoption (among the 104 occupations covered) from 17.8 percent in 2024 to 61.8 percent in 2035. Output nearly triples and its price falls substantially. These gains are labor saving. Productive labor declines from 0.791 to 0.668, while validation labor increases only from 0.008 to 0.026. Consequently, total employment falls by approximately 13 percent and the equilibrium wage falls by 43 percent. The model therefore generates substantial productivity and output gains together with losses in employment and wages because the validation labor channel is quantitatively smaller than the productive labor displacement channel. Because the 104 occupations in the sample account for only 15.25 percent of U.S. salaried employment I conduct an additional national extrapolation exercise, where the remaining 84.75 percent is included as a residual sector with no AI adoption and fixed labor demand at its initial level. Under this extrapolation, output increases by approximately 24 percent, employment falls by 4.9 percent, and the equilibrium wage falls by 18.2 percent between 2024 and 2035. This provides a more attenuated scenario in which a large share of economic activity is either not exposed to AI or does not find it sufficiently cost effective to adopt.

⁴For the rest of the paper the terms task and work activity will be used interchangeably.

The remainder of the paper is organized as follows. Section 2 presents a literature review. Section 3 describes the data. Section 4 presents the model and its equilibrium. Section 5 formalizes and explains the displacement, scale, and validation channels. Section 6 illustrates the model's mechanisms through comparative statics and transition dynamics. Section 7 describes the estimation, and Section 8 reports the quantitative results and conditional forecasts. Section 9 concludes.

2 Relation to the literature

This paper speaks to the broad literature studying how technological change affects labor demand and wages. The literature studying technological change from a task perspective seems to agree that new technologies do not generally replace entire occupations; instead, they substitute labor in some tasks while complementing it in others. Autor et al. (2003) use this distinction to show that computers substituted certain routine activities while increasing demand for workers performing non routine tasks. Acemoglu and Restrepo (2018) study the labor displacement effect of automation and the associated effects on productivity growth and the creation of new labor intensive tasks. In the same lines, Acemoglu and Restrepo (2022) find that automation accounts for a substantial part of the wage reduction experienced by workers specialized in routine tasks.

In their task based framework Acemoglu and Restrepo (2019) and Acemoglu and Autor (2011) decompose the labor market response into a displacement effect and a reinstatement effect. Under that setup capital directly substitutes labor in some tasks while the creation of entirely new tasks reinstates labor. The net effect on the labor share and wages depends on which force dominates. The present paper's labor displacement versus human validation labor creation trade off is structurally related to this decomposition. However, the presented mechanism differs in an important aspect: Acemoglu and Restrepo (2019) "reinstatement" operates by creating genuinely new tasks and job titles for labor to specialize in, whereas the human validation requirement introduced in this paper assigns recurring labor to the same AI automated task, rather than displacing labor into new activities.⁵ This distinction matters for empirical quantification and measurement, because reinstatement effects have historically been slow and difficult to measure, while the validation margin is task specific, is already somewhat observable, and can be disciplined jointly with adoption rates as this paper aims to show.

⁵This paper particularly proposes the distinctive feature that an AI adopted task/workflow can both eliminate productive labor and create a recurring requirement for human validation paid in labor units (time). The equilibrium wage thus depends on which of these two forces is larger. Consequently, neither technical exposure nor a high rate of adoption is sufficient to determine whether workers gain or lose from AI.

The papers closest in spirit to the present quantitative exercise are [Acemoglu \(2024\)](#) and recently [Korinek et al. \(2026\)](#). [Acemoglu \(2024\)](#) provides a back of the envelope calculation of AI's effect on productivity and labor income, treating AI as a task level cost shock. Relative to that exercise, the present paper endogenizes the adoption margin, structurally estimates the AI unit-cost schedule from realized adoption rather than calibrating it, and separates the exposure margin from the validation margin, which allows the model to speak to the sign as well as the magnitude of the wage response. To the best of my knowledge, this paper is the first to introduce a fixed and recurring human validation requirement measured in labor units, inside an endogenous task adoption model. [Korinek et al. \(2026\)](#) present a calibrated task based search and matching model with two broad occupational groups, cognitive and noncognitive. Their model explicitly captures unemployment, occupational reallocation, wage inequality, and the distribution of income between labor and capital. In their simulations, AI capabilities, AI adoption, the subsequent division between task automation and AI augmentation of labor, and the reinstatement of new labor tasks follow exogenously specified scenario paths. Across their modest to extreme simulated scenarios, GDP in 2030 rises by between 1.6 and 32.4% relative to the no AI path, aggregate unemployment increases from 3.8% to between 3.9 and 11.9%, and the labor share falls from 60% to between 45.2 and 59.4%. In contrast, I endogenize task level AI adoption throughout the transition and emphasize the role of human validation, a fixed, recurring time cost that is not productive at the margin, unlike AI augmentation. I structurally estimate the magnitude of the validation requirement and discipline the AI unit cost schedule using observed adoption rates.

The paper also relates to the quantitative literature that connects changes in technology prices to labor market outcomes. [Krusell et al. \(2000\)](#) show that the decline in the relative price of equipment can explain the rise in the skill premium when equipment is more complementary with skilled than with unskilled labor. More recently, [Caunedo et al. \(2023\)](#) construct occupation specific measures of capital embodied technical change⁶ and show that differences in capital-labor substitutability are central for explaining occupational employment reallocation. These studies establish two principles useful for the present paper. First, that technological change can be disciplined using the effective price of a productive input rather than an unexplained productivity residual. Second, the labor market effect of a fall in that price depends on the structure of production and not on the price decline alone. Following this approach, the relevant object in this paper is the task specific unit cost of AI services, that is determined jointly by the price of AI

⁶That is productivity improvements that require the renewed capital investment rather than pure direct increases in productivity.

compute and AI productivity.⁷ The model differs from [Caunedo et al. \(2023\)](#) in that AI adoption is a discrete choice between production methods rather than a smooth change in the capital labor ratio. In contrast, the present model abstracts from worker sorting and multiple skill groups, to focus on isolating the aggregate wage mechanism. The relevant heterogeneity in this paper is instead across tasks in AI costs, task specific validation requirements, and setup costs.

The endogenous selection of tasks into AI production also connects the paper to the technology adoption literature. That literature distinguishes the technical availability of a technology from its economically profitable use and emphasizes fixed costs, heterogeneous returns, and declining technology costs as determinants of diffusion, see [Griliches \(1957\)](#), [Mansfield \(1961\)](#) and [Comin and Hobijn \(2010\)](#). In the present framework, AI capability may be available for many tasks, and in the same way as [Griliches \(1957\)](#) and [Gilchrist and Williams \(2000\)](#) putty-clay models, in this model technical availability does not imply immediate adoption: a technology is adopted only when it becomes sufficiently profitable under local economic conditions. Moreover once installed, the production method cannot be freely reversed. In contrast to that literature, the present mechanism does not require physical capital, machine vintages, or endogenous capacity utilization. A one time setup requirement, recurring validation labor, and irreversible task adoption are sufficient to generate staggered adoption and non-monotone transition paths. The framework is also related to [Aghion et al. \(2018\)](#), who show that non automated tasks can become bottlenecks as automation advances. In this model, however, human labor is required not only in the tasks that remain manual but also to validate the output of tasks that have already adopted AI. Moreover, the set of automated tasks is determined endogenously as AI costs decline rather than imposed as an exogenous technological frontier.⁸

The paper relates to economic models that specifically model AI technology as a distinct technology rather than as a generic productivity shifter. [Agrawal et al. \(2019b\)](#) characterize recent advances in AI as reductions in the cost of "prediction", an explicit input into decision making. They show that the labor market consequences are ambiguous: cheaper prediction may automate an entire decision, but it may also increase the demand for complementary human judgment. An argument that motivates the human validation labor mechanism at the center of this paper.

⁷This paper is not the first to take this approach, [Agrawal et al. \(2019a\)](#) frame AI adoption as a result of the fall in the cost of "prediction".

⁸[Acemoglu et al. \(2012\)](#) show that production networks can amplify sector specific technological shocks. The present model abstracts from input-output propagation in order to isolate the comparison between productive labor displaced and validation labor created. Adding production networks would affect the magnitude and propagation of the response but is not necessary to isolate the sign of the wage response in the current setup. I leave the introduction of input-output linkages for future work aiming to capture relevant sectoral composition effects.

Finally, the paper speaks to the rapidly growing empirical literature measuring the economic reach and use of AI. [Eloundou et al. \(2024\)](#) evaluate the share of occupational tasks whose completion time could be substantially reduced by large language models and other complementary software, while [Felten et al. \(2021\)](#) construct an occupation level measure of exposure to advances in artificial intelligence. These measures are informative about technological capability, but they do not explicitly measure whether firms have adopted the technology or how much human labor an operating AI workflow continues to require. [Hampole et al. \(2025\)](#) construct granular measures of firm and occupation level AI exposure and find that a one standard deviation increase in average AI exposure reduces an occupation’s within firm employment share by 5.5–8.0% (OLS specifications) and 10.4–15.0% (IV specifications), over five years. They show that their results are consistent with a static partial equilibrium task model where AI substitutes labor in exposed tasks, but workers’ reallocation toward unaffected tasks and AI induced firm growth potentially offset the decline in labor demand. [Massenkoff and McCrory \(2026\)](#) develop a new measure of observed AI exposure that combines task level AI capabilities with real world usage. Furthermore, using a difference-in-differences strategy, they find no detectable effect on unemployment among workers in highly exposed occupations. However, they find suggestive evidence of slower hiring among younger workers in these occupations. The experimental and workplace studies conducted by [Noy and Zhang \(2023\)](#) and [Brynjolfsson et al. \(2025\)](#) show that generative AI can raise productivity, although the gains vary substantially across tasks and workers. These productivity gains are important inputs into the adoption decision, but by themselves they do not determine the general equilibrium wage response. [Bick et al. \(2026\)](#) document the remarkably rapid diffusion of generative AI using nationally representative surveys of AI use at work and at home. Importantly, they distinguish the availability of generative AI from its realized adoption and intensity of use. In their study [Shao et al. \(2025\)](#) analyze how worker’s assessments, of how much human involvement they would like to retain when AI is used to perform their tasks, combine with experts’ evaluations of the current capabilities of AI agents. Their results reveal substantial heterogeneity in the role that workers continue to play in AI assisted production and show that technical capability does not always align with the desired level of human involvement. This dimension is closely related to the human validation mechanism introduced in the present paper. The present paper embeds these adoption and human involvement margins in a general equilibrium framework in which they jointly determine labor demand and wages.

3 Data

The theory builds on three key necessary measurements from the data: (1) knowing how much human involvement exists in different AI enabled tasks. (2) AI use/adoption by work activities

and occupations. (3) Traditional task and occupation level labor market statistics (employment, wages, working hours). For this purpose I rely on a combination of five publicly available data sources,⁹ (1) WORKBank survey data, (2) O*NET occupation database, (3) the Real Time Population Survey (RPS), (4) the Current Population Survey (CPS) and (5) BLS Occupational Employment and Wage Statistics (OEWS). Finally I make use of data on task exposure to AI,¹⁰ to model its potential effect on AI unit costs. This section provides a brief description of each data source and the relevant information used. The data Appendix G documents in detail the data mergers and construction of the final dataset.

WORKBank. WORKBank surveys U.S. workers about the tasks they perform and how much human involvement they believe those tasks retain when AI is used.¹¹ It complements worker responses with expert assessments of whether the tasks can be automated and how much human agency, domain expertise, and oversight they require. Each occupation and task is already explicitly linked to the corresponding O*NET occupation codes and task ID's classification.¹² The data is collected from January through May 2025 and covers 846 audited tasks in 104 occupations.¹³ After linking the assessments to the current O*NET taxonomy, the sample contains 5,709 worker and 2,048 expert observations. Workers report desired human agency on a five point scale. Experts report human agency, uncertainty, required domain expertise, and the technological capacity for automation, also on a five point scale. The baseline measure of human validation intensity is constructed as the geometric mean of the first three expert ratings.¹⁴ Although these ratings capture relative differences in human validation requirements across occupations and tasks; they do not yet capture such validation in hours. For the estimation of the model I construct relative occupation and task indices and structurally estimate a common scale parameter, Θ , that converts these indices into quarterly validation hours.

O*NET. I use version 31.0 of the O*NET database, the latest released in August 2026.¹⁵ O*NET data provides occupational codes, detailed task statements, task ratings, and a hierarchy connecting Detailed Work Activities (DWAs), Intermediate Work Activities (IWAs), and Generalized Work Activities (GWAs). To balance occupational information detail while maximizing

⁹To the best of my knowledge, detailed workflow records measuring task level AI use, exact time spent on individual tasks, and human review are not publicly available. Labor and AI evaluation platforms such as micro1 and Mercor have pioneered the collection of such information through specialized task oriented data collection workflows. Access to these data would be particularly valuable because it would replace the current self reported proxies with direct measures of task time and validation labor by occupation.

¹⁰Taken from Eloundou et al. (2024).

¹¹WORKBank repository and codebook link [here](#). The link to the project paper [here](#).

¹²I use these identifiers to link the survey responses to the updated ONET 31.0 database.

¹³The worker response file covers 844 of those tasks.

¹⁴The remaining worker human agency and expert human agency measures are left for robustness.

¹⁵O*NET database documentation link: [here](#)

sample size I choose to map IWAs to the task definition described by the theory. This level is detailed enough to retain economically meaningful task variation and, connect to all 104 occupations in the WORKBank sample. I use O*NET relevance and frequency ratings to construct a proxy for the share of an occupation's time assigned to each IWA, combined with CPS occupation employment weights, provides the occupation task weight that is used throughout to construct aggregate measures of adoption, validation labor and productive labor.

Real Time Population Survey. The RPS provides quarterly AI adoption measures through their GenAI Adoption Tracker. The files provide a pooled sample of the 2025 August, November and 2026 February, May waves. To identify variation over time, I use the separately published quarterly RPS series distributed through FRED.¹⁶ The pooled data contains 13,920 individual observations with valid occupation and workplace GenAI use information. The final activity adoption rate is the survey weighted share of performed activities for which a worker reports using generative AI. The occupation use rate is the survey weighted share of workers in an occupation who report using generative AI for their job. To my advantage, the RPS already classify activities using the O*NET IWA taxonomy. The final model estimation uses 194 IWA level adoption rates, one occupation adoption target for each of the 104 occupations, and eight quarterly occupations worker AI use rates from 2024Q3 through 2026Q2 calculated for the selected RPS broad occupations group. Keeping task adoption and worker AI use distinct is important because a worker in a given occupation can use AI in one or several activities.

CPS and auxiliary OEWS data. The CPS provides occupation employment, weekly work hours, and hourly wages. I use the 2024 and 2025 Basic Monthly samples available through IPUMS CPS. The data extract contains 23 monthly samples; October 2025 is absent. Employment calculations use the Basic Monthly final weight and the wage and salary worker subsample. Wage calculations use earnings of eligible outgoing rotation groups with the respective earnings weight. Hourly wages are weekly earnings divided by usual work hours at the main job, trimmed at the weighted national first and ninety ninth percentiles, and expressed in 2025 U.S. dollars. I used May 2024 OEWS employment data shares to merge CPS occupations and classify them using detailed SOC occupations.¹⁷ SOC serves as the common classification that connects CPS occupations to the O*NET occupation hierarchy.¹⁸

¹⁶RPS GenAI Adoption Tracker: [link](#); FRED work use series: [link](#).

¹⁷The BLS National Employment Matrix CPS crosswalk links CPS occupations with SOC occupations, OEWS provides the respective occupational shares.

¹⁸May 2024 OEWS national estimates: [here](#); BLS occupational crosswalks: [here](#); Inflation (CPI) series from FRED: [here](#).

Task exposure to AI. I use AI exposure classifications constructed by [Eloundou et al. \(2024\)](#) to measure the exposure of each O*NET task to large language models. In the original study each task was classified into three categories increasing in exposure.¹⁹ I assign numerical values to these categories and aggregate them using the O*NET hierarchy to get task occupation level measures used for model estimation. This exposure measure is used as a control to shift in the relative AI unit-cost schedule; it is not used as an observed adoption outcome.

Constructed samples. I construct two data sets that are used to estimate the parameters of the model. The first combines WORKBank data with O*NET and contains 1,720 worker occupation task (IWA) observations that encompass the 104 WORKBank occupations. This human validation measurement data set is used to get a reduced form estimate of human validation requirements in the model ($\theta \bar{H}_A$). The second data set is used for the structural estimation of the model. It combines CPS employment, wages and hours, OEWS employment weights, RPS AI adoption by occupation and task targets, the RPS quarterly targets and [Eloundou et al. \(2024\)](#) exposure scores. Section 7 explains the use of this data to estimate the AI unit cost schedule in the model (q_t^{AI}). The final samples come with an important data limitation, the 104 occupations account for 15.25 percent of U.S. salaried employment. Baseline weights are therefore adjusted accordingly; Appendix E reports a national extrapolation that adds the uncovered economy representing the remaining 84.75 percent as a residual sector.

Table 1 reports descriptive statistics for the variables of interest. These statistics are un-weighted and untrimmed across their respective units of observation; the reduced form and structural estimation subsequently applies the occupation and activity weights and trims as described in Appendix G.

¹⁹E0: no meaningful reduction in completion time. E1: an LLM by itself could reduce completion time by at least 50 percent. E2: the reduction would require an LLM combined with additional software or tools.

Table 1: Descriptive statistics for the main estimation inputs

Source	Variable	Mean	Median	Std. dev.	Min.	Max.	N
WORKBank	Worker human agency rating	2.8814	3.0000	1.1443	1.0000	5.0000	5,709
WORKBank	Expert human agency rating	2.6558	3.0000	1.1008	1.0000	5.0000	2,048
WORKBank	Expert uncertainty rating	2.4165	2.0000	1.1029	1.0000	5.0000	2,048
WORKBank	Expert domain expertise rating	3.0781	3.0000	1.1606	1.0000	5.0000	2,048
WORKBank	Expert automation capacity rating	3.4048	3.0000	1.0261	1.0000	5.0000	2,048
WORKBank	Expert validation composite	2.6157	2.6207	0.9193	1.0000	5.0000	2,048
O*NET	Within occupation IWA time weight	0.0605	0.0435	0.0569	0.0032	0.5893	1,720
O*NET	Number of IWAs per occupation	16.5385	16.5000	5.1707	7.0000	30.0000	104
RPS	IWA task adoption rate	0.2389	0.2212	0.1168	0.0233	0.5552	194
RPS	Occupation worker use rate	0.5574	0.5488	0.1545	0.1562	0.8728	104
RPS	Broad occupations quarterly worker use rate	0.4656	0.4666	0.0777	0.3678	0.5634	8
CPS	Real hourly wage (2025 U.S. dollars)	43.8488	34.5319	34.3118	6.8764	234.9986	49,679
CPS	Usual weekly working hours	40.3295	40.0000	9.1805	1.0000	168.0000	221,879
Eloundou et al.	Task AI exposure score	0.5223	0.5000	0.3283	0.0000	1.0000	1,720

Notes: The sample contains 1,720 occupation task pairs, not $104 \times 194 = 20,176$ because each of the 104 occupations is linked to only the O*NET activities relevant to it, rather than to every activity in the data. The 49,679 wage observations come from the smaller CPS outgoing rotation group, where earnings are collected. The 221,879 hours observations come from the broad sample of employed workers.

4 The Model

Time is discrete. There is a representative household, a representative producer, a competitive supplier of AI services.²⁰ A final composite good is produced combining N task outputs indexed by $i \in \{1, \dots, N\}$.

4.1 Representative Household

There is an infinitely lived representative household that derives utility from a composite consumption good c_t , an endowed linear consumption good x_t (the numeraire) and experiences disutility from labor supply L_t . The household maximizes lifetime utility.

$$\max_{x_t, c_t, L_t} \sum_{t=0}^{\infty} \gamma^t [x_t + v(c_t) - \chi(L_t)] \quad \text{s.t.} \quad x_t + p_t(\mathcal{A}_t) c_t + T_t = w_t(\mathcal{A}_t) L_t + B, \quad x_t \geq 0, \quad c_t \geq 0, \quad L_t \geq 0 \quad (1)$$

where w_t is the wage, p_t is the price of the composite good, T_t is a lump-sum subscription fee to use AI technology, and $B > 0$ is the endowment of the numeraire good.²¹ At any point in time the economy is summarized by the aggregate endogenous state of AI adoption $\mathcal{A}_t$.

²⁰For simplicity AI services are perfectly elastically supplied at fixed prices. It is straightforward to generalize the setup to include an endogenous AI service supply.

²¹This sector is needed purely to allow for quasilinear preferences with a marginal utility of income of one. In the calibration I set the size of this sector to be economically negligible. A tiny sector untouched by AI.

Preferences over c are represented by

$$v(c) = \begin{cases} \omega \frac{\eta}{\eta - 1} (c^{(\eta-1)/\eta} - 1) & \eta \neq 1 \\ \omega \log c & \eta = 1 \end{cases} \quad (2)$$

where $\eta > 0$ is the price elasticity of demand for the composite good and $\omega > 0$ the preference weight.

The disutility of labor takes the conventional form

$$\chi(L) = \frac{\kappa}{1 + 1/\nu} L^{1+1/\nu}, \quad \kappa > 0, \quad \nu > 0. \quad (3)$$

with ν as the Frisch elasticity of labor supply and κ governing the level of disutility.

Since utility is linear in x_t and provided $x_t > 0$ the household's optimality conditions separate into two independent decisions.

Demand for the composite good. The first-order condition $v'(c_t) = \omega c_t^{-1/\eta} = p_t$ gives

$$c_t(p_t) = \left(\frac{\omega}{p_t} \right)^\eta \quad (4)$$

When $\eta = 1$, nominal expenditure on the composite is constant at $p_t c_t = \omega$, so every reduction in the composite's price translates one to one into higher quantity demanded.

Labor supply. The first-order condition $w_t = \chi'(L_t)$ gives

$$L_t(w_t) = \left(\frac{w_t}{\kappa} \right)^\nu. \quad (5)$$

Labor supply depends only on the wage and not on household wealth. A change in AI technology that lowers the price of the composite good and raises real income does not, by itself, alter hours supplied. Deliberately this ensures that all equilibrium wage movements are driven by shifts in labor demand, not by income effects on the supply side.

Numeraire consumption. Given optimal L_t and c_t , numeraire consumption is determined (residually) by $x_t = w_t L_t + B - p_t c_t - T_t$

4.2 Production

The final composite is produced from complementary task outputs y_{it} :

$$Y_t = \left[\frac{1}{N} \sum_{i=1}^N y_{it}^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}} \quad 0 < \sigma < 1 \quad (6)$$

Technology alternatives and their adoption. Each task i can be operated using either a manual technology (M) or artificial intelligence (AI). Before technology installation, the task operator is free to choose between the two technologies. Once AI is installed, it cannot be reversed to manual. The installed method is therefore a state variable. Let indicator $a_{it} \in \{0, 1\}$ denote the installed method state, with $a_{it} = 0$ denoting the manual technology; $a_{it} = 1$ denotes AI technology. The adoption set $\mathcal{A}$ defining the state of AI adoption in the economy and respective adoption share β are:

$$\mathcal{A}_t \equiv \{i : a_{it} = 1\} \quad \beta_t \equiv \frac{|\mathcal{A}_t|}{N} \quad (7)$$

Setup labor cost. A new AI installation for task i uses $f_i > 0$ labor hours and costs wf_i . AI adoption is endogenous, the firm chooses AI adoption for task i only if the associated net surplus is positive. The endogenous adoption choice is explained in detail in Section 4.6.

4.3 Manual production method (M)

Under manual technology, the firm uses human labor to produce task output y_{it} . Conditional on operating task i manually, the firm minimizes total cost $TC_i^M(l_{it}; w_t)$:²²

$$\min_{l_{it} \geq 0} w_t l_{it} \quad \text{s.t. } A^M l_{it} \geq y_{it}, \quad \text{with } A^M > 0 \quad (8)$$

Where l_{it} is productive labor, and A^M manual productivity. Specifically, labor is productive at the margin $\partial y_{it} / \partial l_{it} = A^M > 0$. Solving for labor, the total cost of producing y_{it} and the variable unit cost q^M are:

$$l_{it} = \frac{y_{it}}{A^M}, \quad TC_i^M(y_{it}; w_t) = \frac{w_t}{A^M} y_{it}, \quad q_t^M(w_t) = \frac{w_t}{A^M} \quad (9)$$

²²Throughout the presentation of the model productivity values are set constant to highlight the role of the heterogeneous labor saving AI technology. Without altering the core theoretical results, it is straightforward to include heterogeneity in the productivity of Manual and AI tasks, that is $A_i^M > 0$ and $A_i^{AI} > 0$. The quantitative results in Section 7 include this task specific type of heterogeneity.

4.4 Production with Artificial Intelligence (AI)

Under AI technology, the firm uses AI compute z_{it} and human validation/review labor h_{it} to produce task output y_{it} . Production $y_{it} > 0$ takes place only after satisfying minimum task specific human review requirements measured in human labor $h_{it} \geq \theta \bar{h}_i$. Thus, conditional on operating task i with AI and producing $y_{it} > 0$, the firm minimizes total cost $TC_{it}^{AI}(z_{it}; w_t, p^{AI})$:

$$\min_{z_{it}, h_{it} \geq 0} p_i^{AI} z_{it} + w_t h_{it}, \quad \text{s.t.} \quad A^{AI} z_{it} \mathbf{1}\{h_{it} \geq \theta \bar{h}_i\} \geq y_{it}, \quad \text{with } A^{AI} > 0 \quad (10)$$

where z_{it} is the quantity of AI compute purchased at price p_i^{AI} ; h_{it} is human validation labor; A^{AI} is AI productivity; $\theta > 0$ captures the average level of validation labor required and $\bar{h}_i > 0$ the relative validation labor requirement dispersion across tasks i ; to keep this interpretation I normalize $\frac{1}{N} \sum_{i=1}^N \bar{h}_i = \bar{H} = 1$, this guarantees the separation of $\bar{h}_i$ from its level θ .²³

The solution to (10) implies:²⁴

$$z_{it} = \frac{y_{it}}{A^{AI}}, \quad h_{it} = \theta \bar{h}_i, \quad TC_{it}^{AI}(y_{it}; w_t, p_{it}^{AI}) = q_{it}^{AI} y_{it} + w_t \theta \bar{h}_i, \quad q_{it}^{AI} = p_{it}^{AI} / A^{AI} \quad (11)$$

With $q_{it}^{AI} \equiv p_{it}^{AI} / A^{AI}$ as the variable unit cost. Notice the dual cost structure: the variable cost of compute increases proportionally to output and the fixed recurring cost of validating the AI work. For completeness the marginal and average production costs are:

$$MC_{it}^{AI}(y_{it}) = \frac{\partial TC_{it}^{AI}}{\partial y_{it}} = q_{it}^{AI}, \quad AC_{it}^{AI}(y_{it}) = q_{it}^{AI} + \frac{w_t \theta \bar{h}_i}{y_{it}} \quad (12)$$

From (10), (11), (12), see that validation affects the decision to adopt AI, the average cost of operating the task, and aggregate labor demand, but it does not affect the marginal cost of producing additional output once the AI technology is operating.

²³Under the general setup only effective review $h_i^* = \theta \bar{h}_i$ matters. The normalization and decomposition of h_i^* is done only to keep a clean interpretation. Conditional on adoption total validation labor is $\bar{H}_A = \frac{1}{N} \sum_{i \in A} \bar{h}_i \leq 1$.

²⁴Unlike AI compute, validation labor does not increase with the amount produced and clearly validation labor beyond the threshold $\theta \bar{h}_i$ does not increase task output.

4.5 Conditional cost minimization

Given adoption set $\mathcal{A}_t$ and production value Y_t . The conditional cost minimization problem of the representative firm boils down to:

$$\min_{y_{it}} \frac{1}{N} \sum_i q_{it} y_{it} \quad \text{subject to (6),} \quad \text{where: } \forall i \quad q_{it}(w_t; \mathcal{A}_t) = \begin{cases} q_t^M(w_t) & i \notin \mathcal{A}_t \\ q_{it}^{AI} & i \in \mathcal{A}_t \end{cases} \quad (13)$$

Which delivers an optimal unit expenditure function: ²⁵

$$m_{\mathcal{A}}(w_t) = \left[\frac{1}{N} \sum_{i=1}^N q_{it}(w_t; \mathcal{A}_t)^{1-\sigma} \right]^{\frac{1}{1-\sigma}} \quad \text{if } p_{it}^{AI} = p_t^{AI} \Rightarrow m_{\beta}(w_t) = \left[(1-\beta_t) (q_t^M)^{1-\sigma} + \beta_t (q_t^{AI})^{1-\sigma} \right]^{\frac{1}{1-\sigma}} \quad (14)$$

With the share of manual tasks in the variable cost being equal to the elasticity of the composite variable unit cost with respect to the wage:

$$s_M(w_t, \mathcal{A}_t) = \frac{\partial \log m_{\mathcal{A}}(w_t)}{\partial \log w_t} = (1-\beta_t) \left(\frac{q_t^M}{m_{\mathcal{A}}(w_t)} \right)^{1-\sigma} \quad (15)$$

Lemma 1 (Conditional task demand). *Given adoption set $\mathcal{A}$, and output Y , cost minimizing task output is*

$$y_{it} = Y_t \left(\frac{m_{\mathcal{A}}(w_t)}{q_i(w_t; \mathcal{A}_t)} \right)^{\sigma} \quad (16)$$

Proof. See Appendix A.1 □

The remaining endogenous aggregates are as follows: From (9) and (11) aggregate composite good demand, productive labor demand and demand for AI compute are:

$$Y_t(w_t) = \omega [m_{\beta}(w_t)]^{-\eta}, \quad L_t^M(w_t) = (1-\beta_t) \frac{Y_t(w_t)}{A^M} \left[\frac{m_{\beta}(w_t)}{q_t^M} \right]^{\sigma}, \quad Z(w_t) = (1-\beta_t) \frac{Y_t(w_t)}{A^{AI}} \left[\frac{m_{\beta}(w_t)}{q_t^{AI}} \right]^{\sigma} \quad (17)$$

Importantly production and productive labor demand depend on AI adoption only through β_t . However specific task AI adoption matters directly for validation labor and setup labor because

²⁵See complete solution to the minimization problem in Appendix A.1

$\bar{h}_i$ and f_i are task specific, where $\mathcal{N}_t = \mathcal{A}_t \setminus \mathcal{A}_{t-1}$ is the set of newly installed tasks.²⁶

$$H_t(\mathcal{A}_t, \theta) = \theta \bar{H}_{\mathcal{A}_t} = \frac{\theta}{N} \sum_{i \in \mathcal{A}_t} \bar{h}_i \quad F_t = \frac{1}{N} \sum_{i \in \mathcal{N}_t} f_i \quad (18)$$

Aggregate labor demand is thus:

$$L_t^D(w_t; \mathcal{A}_t, \theta) = L_t^M(w_t) + H_t(\mathcal{A}_t, \theta) + F_t \quad (19)$$

The equilibrium wage w_t clears the market $L_t(w_t) = L_t^D(w_t; \mathcal{A}_t, \theta)$ Moreover, conditional on a fixed adoption set the wage increases with θ

$$\frac{\partial w_t(\mathcal{A}_t, \theta)}{\partial \theta} = \frac{\bar{H}_{\mathcal{A}_t}}{L_t'(w_t) - L_t^{M'}(w_t)} > 0 \quad (20)$$

Profits and subscription fee. The household pays aggregate $T_t = w_t \theta \bar{H}_{\mathcal{A}_t}$ to the producer as a subscription charge to cover for the validation cost, ensuring the producers profits are zero.²⁷ When making the adoption decision, the producer takes T_t as a given ex-post cost recovery transfer.

$$\Pi_t = p_t Y_t + T_t - w_t \underbrace{(L_t^M + H_t + F_t)}_{=L_t^D} - p_t^{AI} Z_t \quad (21)$$

Marginal cost pricing implies $p_t Y_t = w_t L_t^M + p_t^{AI} Z_t$ that covers manual labor, but not validation and setup labor, setting $T_t = w_t H_t = w_t \theta \bar{H}_{\mathcal{A}_t}$ makes $\Pi_t = 0$.

4.6 Endogenous Adoption

AI technology adoption itself is endogenous and decentralized, the price taking firm evaluates adoption based on whether the new technology increases the surplus at the moment of evaluating installation. This choice is irreversible, meaning that once setup costs are paid and AI is adopted for a given task, it can no longer be produced using the manual technology.

4.7 Perceived operating surplus of AI adoption

The firm takes the aggregate state $(w, P, Y, \mathcal{A})$ as given at the point of considering task i for AI installation. The task output $\tilde{y}_{it}^M(w_t, \mathcal{A}_t)$ it would be required to deliver $Y_{t,\mathcal{A}}$ under the manual

²⁶In steady state the set $\mathcal{N}_t$ is empty because there are no new tasks that adopt AI thus $F_t = 0$

²⁷This is a mere accounting mechanism to allow the producer to cover its fixed validation cost while continuing to charge the marginal cost p_t . T_t has no effect on aggregate resources or welfare.

method is:

$$\tilde{y}_{it}^M(w_t, \mathcal{A}_t) = Y_{t\mathcal{A}}(w_t) \left[\frac{m_{\mathcal{A}}(w_t)}{w_t/A^M} \right]^\sigma \quad (22)$$

Producing this amount manually costs $(w_t/A^M)\tilde{y}_{it}^M$. Producing the same required amount with AI costs $q_i^{AI}\tilde{y}_{it}^M + w_t\theta\bar{h}_i$. The perceived per period operating savings of adoption are therefore:

$$\xi_{it}(w_t, \mathcal{A}_t; \theta) = \left(\frac{w_t}{A^M} - q_i^{AI} \right) \tilde{y}_{it}^M(w_t, \mathcal{A}_t) - w_t\theta\bar{h}_i. \quad (23)$$

The subscription fee introduced in Section 4.5 finances the underlying review wage bill, therefore does not change this real cost comparison.

With $r = \gamma/(1 - \gamma) > 0$ as the implied interest rate. Its perceived present value keeping future prices constant is, ξ_{it}/r . Installation uses $f_i > 0$ labor hours and costs $w_t f_i$. Thus net capitalized surplus is:

$$\Upsilon_{it}(w_t, \mathcal{A}_t; \theta) = \frac{\xi_{it}(w_t, \mathcal{A}_t; \theta)}{r} - w_t f_i. \quad (24)$$

An eligible manual task switches to AI if and only if satisfies the adoption rule $\Upsilon_{it} \geq 0$. Accordingly define the adoption mapping:

$$\Gamma(\mathcal{A}_t) = \{i : \Upsilon_{it}(w_t, \mathcal{A}_t; \theta) \geq 0\}$$

Then, the set of new installations as per the adoption rule is $\mathcal{N}_{t+1} = \{i \notin \mathcal{A}_t : \Upsilon_{it} \geq 0\}$. Then the adoption set evolves as the following law of motion:

$$\mathcal{A}_{t+1} = \mathcal{A}_t \cup \mathcal{N}_{t+1} \quad (25)$$

4.8 Equilibrium

4.8.1 Stationary Competitive Adoption Equilibrium

Definition 1 (Stationary competitive adoption equilibrium). *A stationary competitive equilibrium is a collection of prices, transfers, and quantities $(w, p, T; x, c, Y, L, Z, \{y_i, l_i, z_i, h_i\}_{i=1}^N)$ and an adoption measure set $\mathcal{A}$, such that:*

1. *Given w, p , and the lump sum validation access fee T , the household chosen quantities (x, c, L) maximize its utility subject to its budget constraint.*

2. Conditional on $\mathcal{A}$, the inputs (y_i, l_i, z_i, h_i) minimize the cost of producing Y .
3. Every task operated with AI satisfies its validation constraint, $h_i = \theta \bar{h}_i$, $i \in \mathcal{A}$.
4. Competition implies marginal cost pricing $p = m_{\mathcal{A}}(w)$ and the validation access fee covers the fixed validation wage bill, $T = w\theta \bar{H}_{\mathcal{A}}$ so that the production sector earns zero profits.
5. The goods markets and labor market clear:

$$\begin{aligned} c &= Y \\ L(w) &= L^M(w, \mathcal{A}) + \theta \bar{H}_{\mathcal{A}} \\ x + p^{AI} Z &= B \end{aligned} \tag{26}$$

6. A stationary adoption is induced by the equilibrium adoption rule:

$$\mathcal{A} = \Gamma(\mathcal{A}) \tag{27}$$

4.8.2 Existence and Uniqueness

Lemma 2 (Manual labor demand is strictly decreasing on w). For a fixed adoption set containing at least one manual task:

$$\frac{\partial \log L^M(w, \mathcal{A})}{\partial \log w} = (\sigma - \eta) s_M(w, \mathcal{A}) - \sigma < 0. \tag{28}$$

Proof. See Appendix [A.2](#) □

Proposition 1 (Unique conditional equilibrium wage). For any nonempty feasible production system and fixed $(\mathcal{A}, \theta)$, equation (26) has a unique positive solution.

Proof. Define

$$G(w; \mathcal{A}, \theta) = L(w) - L^M(w, \mathcal{A}) - \theta \bar{H}_{\mathcal{A}}. \tag{29}$$

Labor supply is continuous and strictly increasing; manual labor demand is continuous and strictly decreasing; and review labor is independent of w . Hence G is strictly increasing. At a sufficiently small wage, labor supply is arbitrarily small while either manual or review labor demand is positive, so $G < 0$. At a sufficiently large wage, labor supply diverges while manual labor demand vanishes and review demand remains finite, so $G > 0$. Continuity gives existence and strict monotonicity gives uniqueness. □

4.8.3 Non-Stationary Competitive Adoption Equilibrium

Definition 2 (Non-stationary competitive adoption equilibrium). *Given an initial adoption set $(\mathcal{A}_0, \mathcal{N}_0)$ a non-stationary competitive adoption equilibrium is a sequence of AI adoption measures, prices, transfers, quantities and a measures $\{\mathcal{A}_t, \mathcal{N}_t; w_t, p_t, T_t; x_t, c_t, Y_t, L_t, Z_t, F_t, \{y_{it}, l_{it}, z_{it}, h_{it}\}_{i=1}^N\}_{t \geq 0}$ such that, $\forall t \geq 0$:*

1. *Given prices (w_t, p_t) and fee T_t , the household choice of (x_t, c_t, L_t) solves its maximization problem.*
2. *Conditional on $\mathcal{A}_t$, task inputs $(y_{it}, l_{it}, z_{it}, h_{it})$ minimize production costs and AI tasks satisfy their validation constraints.*
3. *Competitive marginal cost pricing and fee payments satisfy zero profit decentralization.*
4. *The goods and labor markets clear*

$$\begin{aligned} c_t &= Y_t \\ L_t(w_t) &= L_t^M(w_t, \mathcal{A}_t) + \theta \bar{H}_{\mathcal{A}_t} + F_t \\ x_t + p_t^{AI} Z_t &= B \end{aligned}$$

5. *Newly installed tasks satisfies the adoption rule $\mathcal{N}_{t+1} = \{i \notin \mathcal{A}_t : \Upsilon_{it} \geq 0\}$*
6. *The adoption set evolves according to $\mathcal{A}_{t+1} = \mathcal{A}_t \cup \mathcal{N}_{t+1}$*

A stationary competitive adoption equilibrium is the limiting case of the non-stationary competitive adoption equilibrium defined above in which adoption is stationary thus $F_t = 0$.

5 Mechanisms

5.1 The Pre-AI Economy

In this section I define and characterize a Pre-AI Economy, useful to benchmark the effects of AI adoption.

Naturally before AI, the adoption set $\mathcal{A}_0 = \emptyset$ and $\mathcal{N}_0 = \emptyset$. Then

$$m_0(w) = p_0(w) = q_0^M(w) = \frac{w}{A^M} \quad (30)$$

$$Y_0(w) = \omega \left(\frac{w}{A^M} \right)^{-\eta} \quad (31)$$

$$L_0^D(w) = \frac{Y_0(w)}{A^M} = \omega (A^M)^{\eta-1} w^{-\eta} \quad (32)$$

Equating labor supply (5) and labor demand (32) gives the pre-AI equilibrium wage:

$$w_0 = \left[\omega (A^M)^{\eta-1} \kappa^\nu \right]^{\frac{1}{\nu+\eta}} \quad (33)$$

5.2 Productive labor, validation labor, and the wage response

AI affects labor demand through two distinct channels. First, automation eliminates productive labor from the tasks that adopt AI. At the same time, the associated decline in the price of the composite good expands output and increases demand for the tasks that remain manual. Consequently, even though AI is labor saving within every adopted task, it need not reduce aggregate productive labor. Second, every adopted task creates a fixed demand for validation labor. The equilibrium wage and thus total labor income depends on the combined strength of these two channels.

5.2.1 The response of productive labor to AI adoption

Assume a productivity improving AI technology shock that makes $q^{AI} < q_0^M$ and results in a fixed adoption set $\mathcal{A}$ with adoption share β . After adoption the proportional change in the variable unit cost of the composite good, evaluated at w_0 , is

$$\varrho_\beta \equiv \frac{m_\beta(w_0)}{q_0^M(w_0)} \in (0, 1) \quad (34)$$

Productive labor demand after adoption, evaluated at the initial wage, is

$$L_\beta^M(w_0) = (1 - \beta) \frac{Y_\beta(w_0)}{A^M} \left[\frac{m_\beta(w_0)}{q_0^M(w_0)} \right]^\sigma \quad (35)$$

And relative to the pre-AI productive labor demand.

$$\frac{L_\beta^M(w_0)}{L_0^D(w_0)} = (1 - \beta) \frac{Y_\beta(w_0)}{Y_0} \left[\frac{m_\beta(w_0)}{q_0^M} \right]^\sigma = (1 - \beta) \varrho_\beta^{\sigma - \eta} \quad (36)$$

Theorem 3 (Productive labor displacement and the scale effect). *For a fixed adoption share $\beta \in (0, 1)$, it follows from the ratio (36) that:*

1. *If $\eta \leq \sigma$, then*

$$L_\beta^M(w_0) < L_0^D(w_0). \quad (37)$$

Automation necessarily reduces aggregate productive labor demand at the initial wage.

2. *If $\eta > \sigma$, then*

$$L_\beta^M(w_0) < L_0^D(w_0) \iff \varrho_\beta > (1 - \beta)^{1/(\eta - \sigma)}, \quad (38)$$

$$L_\beta^M(w_0) = L_0^D(w_0) \iff \varrho_\beta = (1 - \beta)^{1/(\eta - \sigma)}, \quad (39)$$

$$L_\beta^M(w_0) > L_0^D(w_0) \iff \varrho_\beta < (1 - \beta)^{1/(\eta - \sigma)}. \quad (40)$$

Thus, when demand is sufficiently elastic relative to task substitutability, a sufficiently large reduction in the composite's variable unit cost raises productive labor demand in the remaining manual tasks above its pre-AI level.

Proof. See Appendix A.3 □

Equation (36) separates the two forces. The term $1 - \beta$ captures direct displacement: fewer tasks continue to employ productive labor. The term $\varrho_\beta^{\sigma - \eta}$ captures the response of productive labor in the tasks that remain manual. A lower composite price expands aggregate demand with elasticity η , while substitution toward cheaper AI tasks is governed by σ . When $\eta > \sigma$, the output expansion can dominate both direct displacement and substitution away from manual tasks. Task complementarity makes this outcome more likely because the remaining manual tasks are still required to produce the composite.

When $\beta = 1$, no manual tasks remain and $L_\beta^M(w_0) = 0$. Hence full automation necessarily eliminates productive labor demand, although it may still create validation labor.

5.2.2 The wage response

Define the change in productive labor displaced by AI, evaluated at the initial wage, as

$$\Delta_{\beta}^M \equiv L_0^D(w_0) - L_{\beta}^M(w_0). \quad (41)$$

The quantity Δ_{β}^M may be positive, zero, or negative.

Theorem 4 (The wage response to AI adoption). *For a nonempty adoption set $\mathcal{A}$*

1. *If $\Delta_{\beta}^M > 0$, AI reduces (substitutes) productive labor demand at the initial wage. Furthermore, there exists a positive threshold*

$$\theta^*(\mathcal{A}) \equiv \frac{\Delta_{\beta}^M}{\bar{H}_{\mathcal{A}}} = \frac{L_0^D(w_0) - L_{\beta}^M(w_0)}{\bar{H}_{\mathcal{A}}} \quad (42)$$

Such that the equilibrium wage under AI adoption satisfies

$$\theta < \theta^*(\mathcal{A}) \implies w(\mathcal{A}, \theta) < w_0 \quad (43)$$

$$\theta = \theta^*(\mathcal{A}) \implies w(\mathcal{A}, \theta) = w_0 \quad (44)$$

$$\theta > \theta^*(\mathcal{A}) \implies w(\mathcal{A}, \theta) > w_0 \quad (45)$$

2. *If $\Delta_{\beta}^M = 0$, the output expansion exactly offsets productive labor displacement. The wage is unchanged when $\theta = 0$, while at any positive validation requirement $\theta > 0$ the equilibrium wage increases $w(\mathcal{A}, \theta) > w_0$.*
3. *If $\Delta_{\beta}^M < 0$, AI raises productive labor demand in the remaining manual tasks above their pre-AI level. The productive labor channel alone raises the wage, and validation labor reinforces the effect increasing the wage $w(\mathcal{A}, \theta) > w_0$ for every $\theta \geq 0$*

Proof. See Appendix A.5 □

The two cases have a distinct economic interpretation. When $\Delta_{\beta}^M > 0$, AI is aggregate productive-labor saving. The wage then depends on whether the validation labor created by AI is sufficient to replace the productive hours displaced. When $\Delta_{\beta}^M \leq 0$, the expansion in output and demand for the remaining complementary manual tasks already prevents the wage from falling. Validation labor is no longer required to reverse the declining wage effect; it only strengthens an increase that is already generated by the productive labor channel.

Notice that the above holds regardless of the intensity of AI adoption β , as long as $\beta > 0$. To see this decompose:

$$H(\mathcal{A}, \theta) = \theta \underbrace{\beta}_{=\beta} \underbrace{\bar{h}_{\mathcal{A}}}_{=\bar{h}_{\mathcal{A}}} = \theta \underbrace{\frac{|\mathcal{A}|}{N} \frac{1}{|\mathcal{A}|} \sum_{i \in \mathcal{A}} \bar{h}_i}_{\bar{h}_{\mathcal{A}}}$$

Corollary 1 (Aggregate employment response). *Equilibrium employment responds in the same direction as the equilibrium wage, consequently, all three cases established in Theorem 4 for the equilibrium wage also apply to equilibrium employment.*

Proof. See Appendix A.4 □

Wealth effects of AI Adoption. Under the current setup, aggregate labor demand is downward sloping in the wage. AI adoption shifts the labor demand schedule, while the labor supply schedule remains unchanged, thus equilibrium moves along the upward sloping labor supply curve, so the equilibrium employment and the equilibrium wage move in the same direction. Introducing positive wealth effects of AI on leisure would generally shift labor supply pushing wages upward. However a wealth effect translated to consumption of complementary normal goods, could raise productive labor demand thus reducing $\theta_W^*(\mathcal{A})$. Consequently, a smaller amount of validation labor could be sufficient to prevent the wage from falling. Wealth effects therefore could counteract the wage reducing effect of productive labor displacement. See Appendix C for the extension. Quantifying the magnitude of the wealth effects remains an empirical question left for future extensions.

6 Numerical simulations

This section illustrates the main mechanisms through numerical simulations of a pseudo calibrated model.²⁸ It first presents a set of comparative statics of steady state solutions for different model parametrizations. Finally it presents simulated transition dynamics from a Pre-AI economy to an AI adoption Economy. The simulations showcase the following effects associated to AI adoption:

1. **Productive labor displacement:** It occurs when adopting AI removes the direct productive/manual labor used to produce a task, thus shifting labor demand down and thus wages.

²⁸See the chosen parameter values in Appendix B.2

2. **Fixed validation labor creation:** Regardless of the level of adoption, every adopted task requires $\theta \bar{h}_i$ review hours. This shifts labor demand upward and thus equilibrium wages.
3. **Slowing labor displacement via complementarity $\sigma_T < 1$:** As AI reduces the cost of adopted tasks, the relative importance of remaining manual task raises. Thus limiting the decline in productive labor and can raise output from to manual tasks.
4. **Adoption feedback effect:** Adopting AI increases human validation labor $H(\mathcal{A}, \theta)$ and wages, but at the same time making marginal tasks less attractive to automate. This is why $w(\theta)$ can eventually turn downward even though $\partial w / \partial \theta > 0$ is conditional on a fixed adoption set.
5. **Temporary setup congestion:** During AI installation, F_t raises labor demand and can temporarily lift wages while crowding labor out of production. This channel vanishes in the stationary allocation.

6.1 Comparative Statics

Increasing human validation intensity through θ . Increasing human validation intensity requirements carries both the fixed validation labor creation effect and the adoption feedback effect. Figure 4 shows the simulation exercise for values of $\theta = [0.1, 2.0]$. See from panel A, that wages increase as $H(\mathcal{A}, \theta)$ increases. At $\theta = 1.2$ the adoption feedback effect kicks in. By making adoption less attractive, validation labor decreases and it is substituted by lower equilibrium level of productive labor, thus reducing wages until it converges back to the a Pre-AI economy with zero adoption.

Increasing AI compute price p_{AI} . An increase in AI compute cost results into two opposing effects. First, is the standard scale effect. More expensive AI raises the variable unit cost q^{AI} , which raises the price of the composite good and reduces output. Because validation labor $H(\mathcal{A}, \theta)$ remains constant as long as adoption is unchanged, total labor demand falls, lowering the equilibrium wage. Second, this is a reverse displacement effect, as AI costs rise, adoption naturally decreases, increasing demand for productive labor and increasing wages.²⁹ Figure 5 and Figure 10, show the respective comparative statics exercise for economies where for values of $p^{AI} = [0.5, 5.0]$ with a low $\theta = 0.2$ and high $\theta = 0.95$, respectively. The discrete jump down of

²⁹For this calibration output goes down because manual labor is less productive than AI technology.

labor demand and increase in output as adoption increases are a reflection the discrete adoption decision rule. The same jumps occur at the task level. ³⁰

6.2 Transition to an AI economy

This section simulates transition dynamics for two identical economies that start from the same Pre-AI economy, in case of AI adoption, they differ only on the required amount of human validation labor required per task θ . To trigger adoption, both economies are shocked with the same gradual 90% decline of AI costs $\bar{q}_t^{AI}$.³¹ The exercise shows that depending only on the required amount of human validation labor $H(\mathcal{A}, \theta)$ the two economies converge to different steady states, one with higher equilibrium employment and wages whereas the other to an equilibrium with lower employment and wages. This divergence is not caused by any differences in AI adoption, because both economies converge to the same endogenous level of AI adoption $\beta_{ss} = 0.83$. Table 2 presents the steady states comparison between the two economies and Figure 2 and Figure 3 presents a comparison of their transition paths.

Figure 2 Panel (a) and (d) show that equilibrium employment moves in the same direction as the wage. With low validation intensity, the net reduction in labor demand converges to a new steady state wage below its pre-AI level. With high validation intensity, respective high validation labor requirements lead to a wage above the pre-AI level, this occurs despite both converging to the same level of AI adoption as shown in Panel (b). Note that, different adoption paths arise endogenously because adoption conditions in each economy differ long the transition, the high θ adopts AI later because a high θ makes its validation requirement more expensive. The price, AI compute demand and output panels show the respective decline in production costs, increasing AI input use and increasing output.

During the transition every newly adopted task adds recurring validation labor and temporary setup labor. Wage pressure therefore rises sharply in the large adoption quarters, creating wage spikes. In the next quarters, fewer tasks install, setup labor falls, and the labor saving and output price effects dominate. These wage spikes are evident in the transition path of the economy with high θ . The decomposition of total labor demand is presented in panels (c) and (d) of Figure 3.

Figure 3 panel (a) and (b) show that as AI becomes economically viable for a larger pool of tasks, endogenous adoption lowers production costs thus expanding output. The labor share naturally declines more in the faster adopting economy due to lower adoption payroll costs.

³⁰Since $q^{AI} = p^{AI}/A^{AI}$, a proportional price increase and the corresponding productivity decrease generate identical equilibrium allocations if they imply the q^{AI} . Only physical compute use Z , differs; compute expenditure, $p^{AI}Z$, is unchanged.

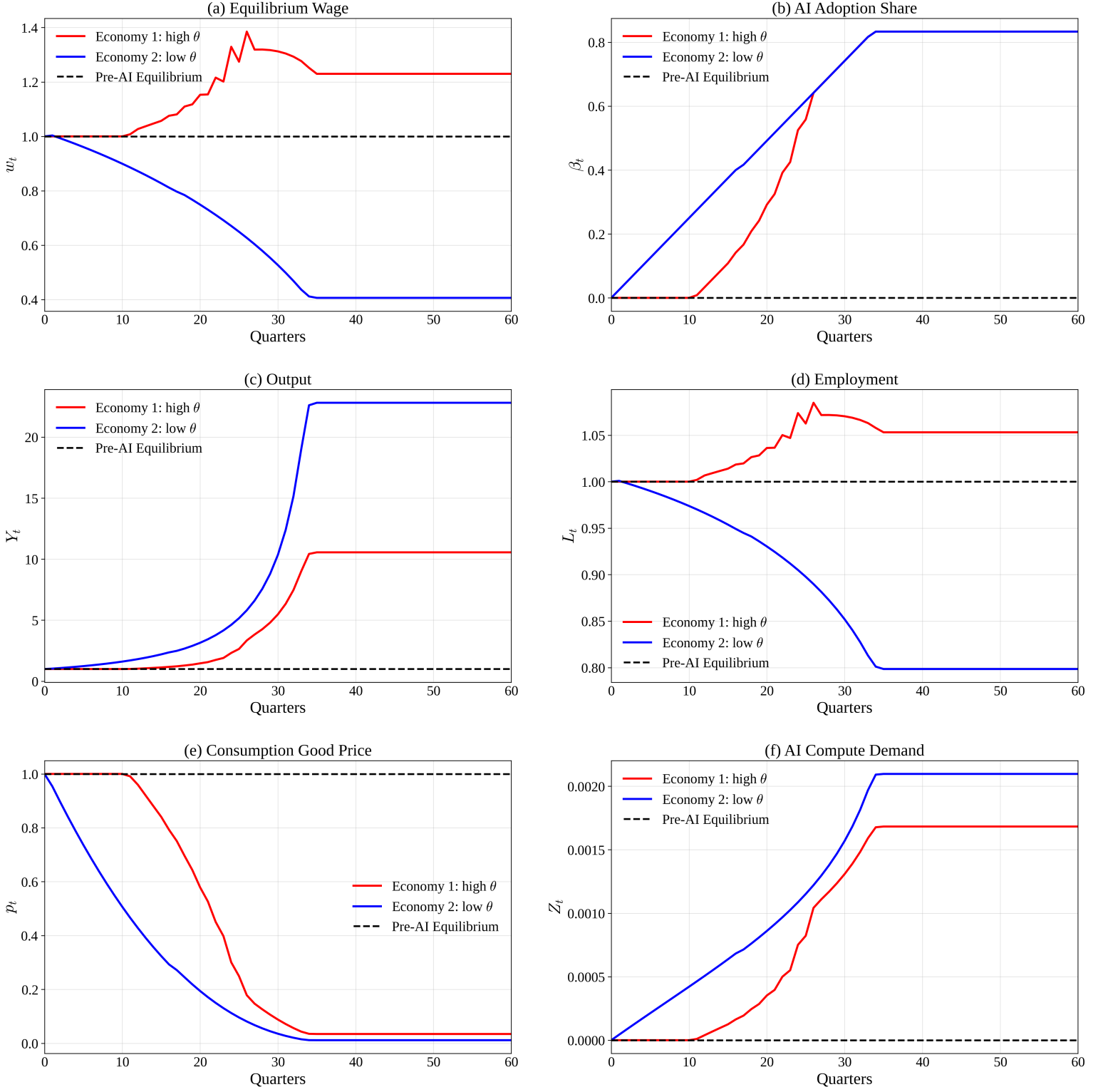
³¹Figure 8 plots the evolution of the exogenous AI unit costs.

Panels (c) and (d) show the labor market response and its decomposition in both economies: both experience displacement of productive labor and an increase in the validation labor required for AI adopted tasks. Although small, the contribution of the one time setup labor costs during installation is present until convergence.

Table 2: Steady state comparison between two economies after AI adoption

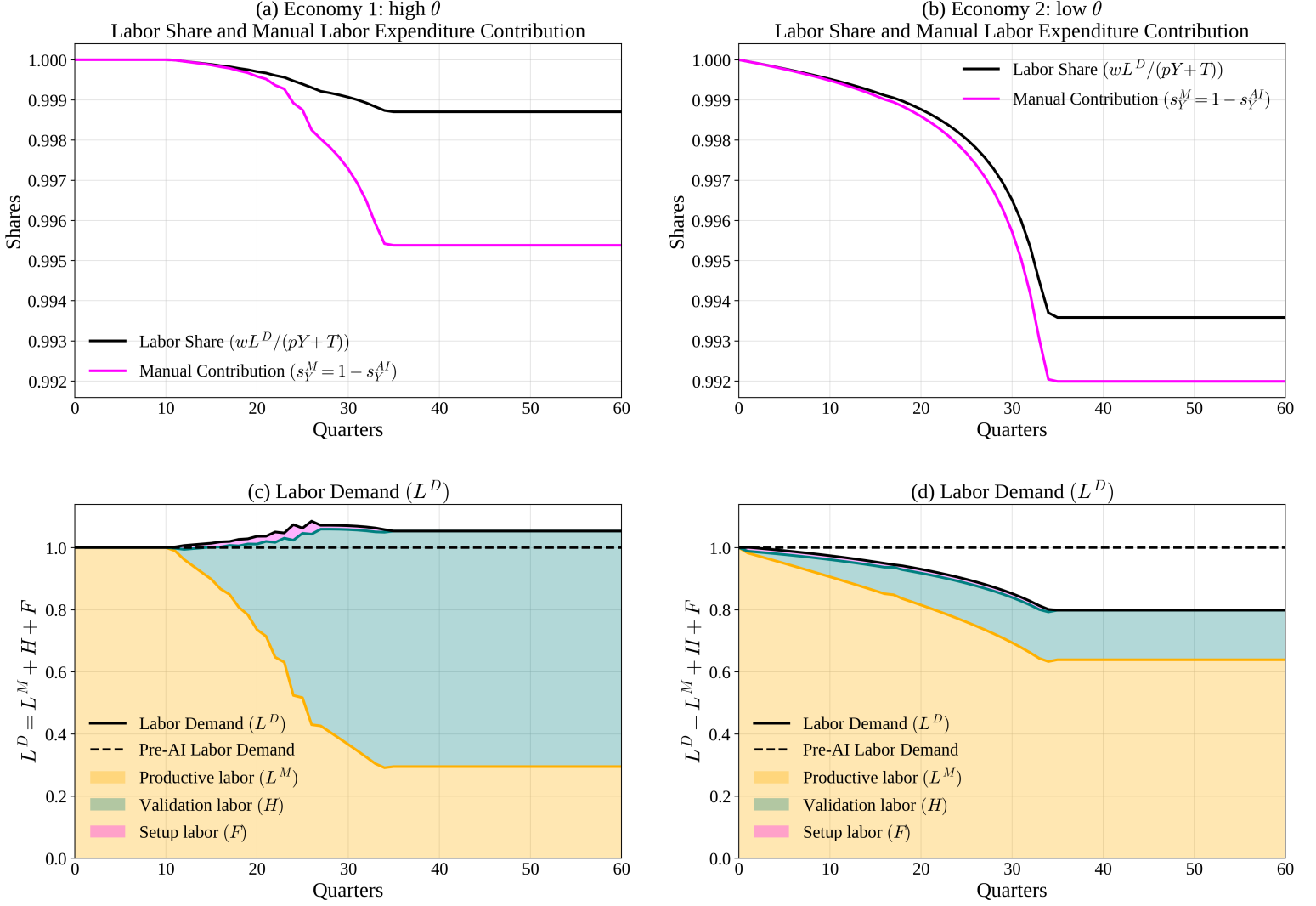
	Pre-AI	Economy 1 High validation θ	Economy 2 Low validation θ
Human Validation intensity θ	—	0.95	0.20
Quarter of first endogenous AI adoption	—	11	1
Endogenous AI adoption share β	0	0.833	0.833
Validation base $\bar{H}_{\mathcal{A}^*}$	0	0.798	0.798
Conditional threshold $\theta^*(\mathcal{A}^*)$	—	0.826	0.826
Equilibrium wage w	1.000	1.230	0.406
Wage change	—	+23.0%	−59.4%
Productive labor L^M	1.000	0.295	0.639
Human Validation labor $H = \theta \bar{H}_{\mathcal{A}^*}$	0	0.758	0.160
Total labor demand $L^D = L^M + H$	1.000	1.053	0.798
Labor share	1.000	0.999	0.994
Composite good price p	1.000	0.0345	0.0115
Composite output Y	1.000	10.560	22.813
Manual contribution s_Y^M	1.000	0.995	0.992
AI contribution s_Y^{AI}	0	0.005	0.008
AI compute Z	0	0.00168	0.00210
Numeraire-sector share	0.220%	0.601%	0.833%

Figure 2: Aggregate transition dynamics



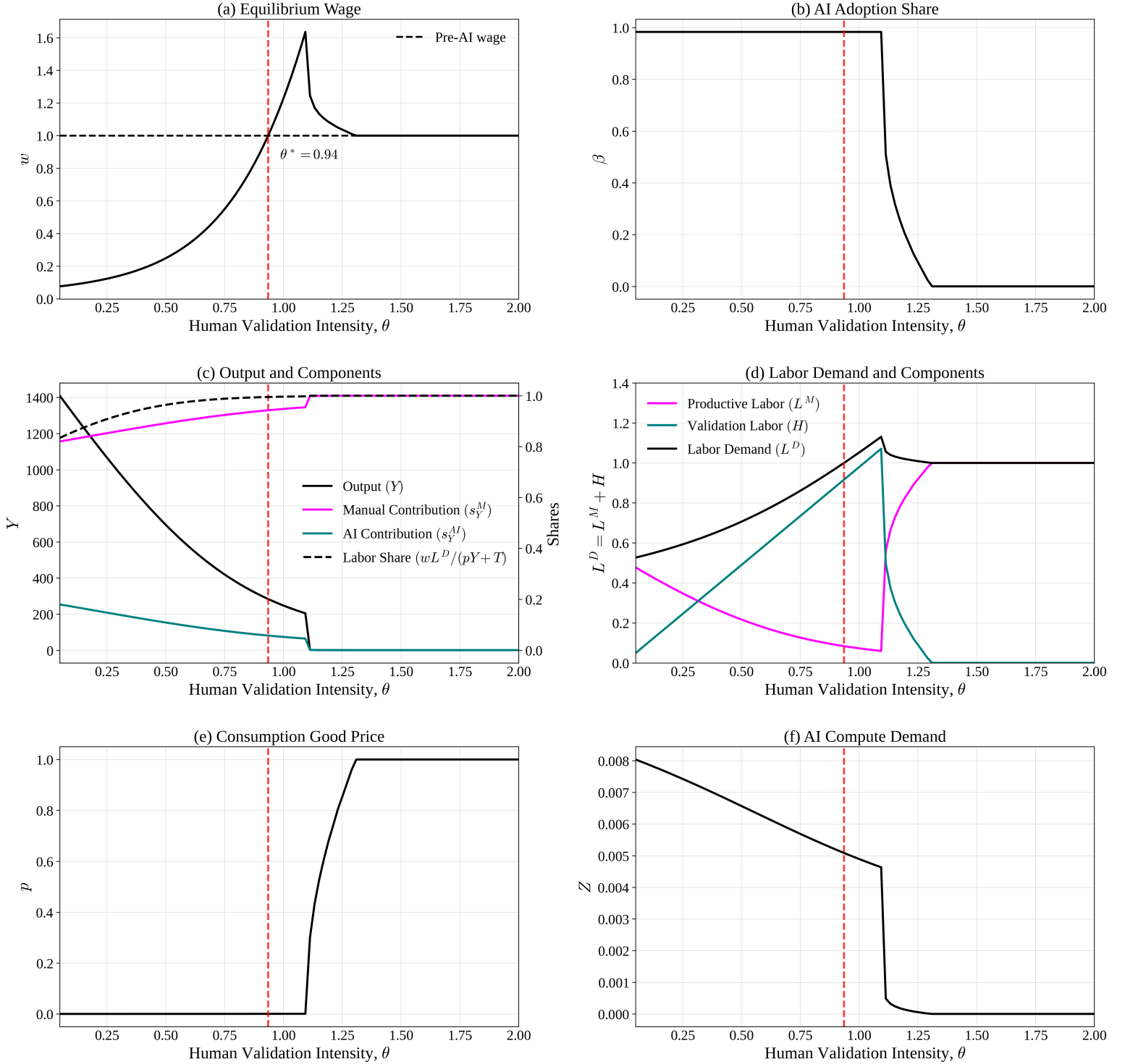
Notes: The parametrization is $\theta^{High} = 0.95$, $\theta^{Low} = 0.2$, $N = 120$, $\sigma = 0.50$, $A^M = 1$, $\eta = 0.70$, $\omega = 1$, $\nu = 0.25$, $\kappa = 1$, $p^{AI} = 1$, $Nh_i = 1$, $f_i = 0.50$, $r = 0.04$, $q_0^{Pre-AI} = 5$, $B = 0.0022$. Simulations run for 80 periods.

Figure 3: Labor market transition dynamics



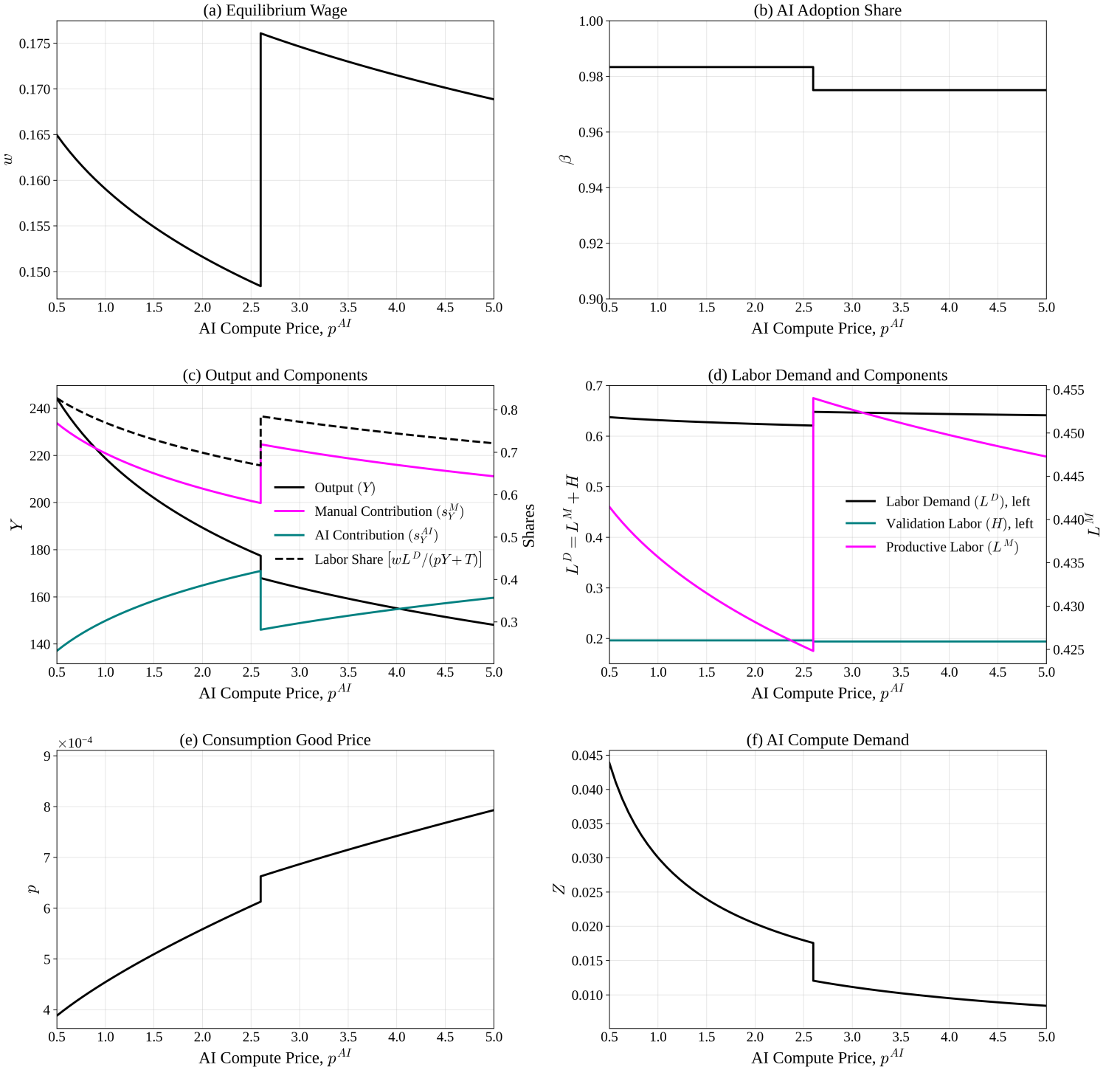
Notes: The parametrization is $\theta^{High} = 0.95$, $\theta^{Low} = 0.2$, $N = 120$, $\sigma = 0.50$, $A^M = 1$, $\eta = 0.70$, $\omega = 1$, $\nu = 0.25$, $\kappa = 1$, $p^{AI} = 1$, $Nh_i = 1$, $f_i = 0.50$, $r = 0.04$, $q_0^{Pre-AI} = 5$, $B = 0.0022$. Simulations run for 80 periods.

Figure 4: Increasing required validation labor level through θ



Notes: The parametrization is $N = 120$, $\sigma = 0.50$, $A^M = 1$, $\eta = 0.70$, $\omega = 1$, $\nu = 0.25$, $\kappa = 1$, $p^{AI} = 1$, $\bar{q}^{AI} = 0.033$, $Nh_i = 1$, $f_i = 0.50$, $r = 0.04$, $q_0^{Pre-AI} = 5$, $B = 0.01694$. $\bar{A}^{AI}$ is set to arbitrary high number. Simulations run for 80 periods, only steady state values reported.

Figure 5: Increasing AI compute price, $\theta = 0.2$



Notes: The parametrization is $N = 120$, $\sigma = 0.50$, $A^M = 1$, $\eta = 0.70$, $\omega = 1$, $\nu = 0.25$, $\kappa = 1$, $\theta = 0.20$, $Nh_i = 1$, $f_i = 0.50$, $r = 0.04$, $q_0^{Pre-AI} = 5$, $B = 0.045$. $\bar{A}^{AI}$ is set to arbitrary high number. Simulations run for 80 periods, only steady state values are reported.

7 Estimation

The model is estimated using the U.S. 2024-2025 combined data set described in Section 3 and Appendix G. The model is estimated at the $k = (i, g)$, task $i \in \mathcal{I}$, occupation $g \in \mathcal{G}$ level, using constructed task-occupation weights π_{ig} .³² The estimation proceeds in two steps: the first stage concerns the reduced form estimation of the human validation requirements $\theta \bar{h}_i$. The second stage concerns the structural estimation of q_{it}^{AI} .

Stage 1: Estimating human validation requirements $\theta \bar{h}_i$. I estimate the following reduced form relation:

$$\log v_{nig} = \mu + \alpha_g + \lambda_i + \varepsilon_{nig} \quad (46)$$

Where v_{nig} , is the constructed human validation proxy by worker n for task i in occupation g , $\alpha_g = \log \tilde{\theta}_g$ and $\lambda_i = \log \tilde{h}_i$ capture the occupation and task effects respectively. Then, the normalized validation parameters such that $\sum_i \pi_i \bar{h}_i = 1$, are recovered by normalizing:

$$\tilde{\theta}_g = \frac{\exp(\alpha_g)}{\prod_g \exp(\alpha_g)^{\pi_g}}, \quad \bar{h}_i = \frac{\exp(\lambda_i)}{(1/\mathcal{I}) \sum_i \exp(\lambda_i)} \quad (47)$$

At this stage, $\tilde{\theta}_g \bar{h}_i$ is a relative index rather than a quantity measured in labor hours. To anchor the aggregate scale to labor hours units, define $\Theta = \theta_g / \tilde{\theta}_g$. Θ is estimated in the next stage and disciplined to match 2024Q3-2025Q4 occupation level adoption rates. Notice that Θ is already the average required validation labor outlined in (11). To see this, define the validation base associated to an adoption set $\mathcal{A}$ as $\bar{H}_{\mathcal{A}} = \sum_{(g,i) \in \mathcal{A}} \pi_{gi} \tilde{\theta}_g \bar{h}_i$, then the total validation labor is $H(\mathcal{A}, \Theta) = \Theta \bar{H}_{\mathcal{A}}$.

Stage 2: Structural estimation of the cost schedule q_{it}^{AI} . The estimation is conducted via indirect inference using a simulated method of moments methodology.³³ The full cost schedule q_{igt}^{AI} is disciplined to match the 2024Q3 - 2026Q2 task adoption rates. It is parametrized as

³²Denote the respective marginals π_g and π_i and conditional $\pi_{i|g}$, thus $\pi_{ig} = \pi_g \pi_{i|g}$. See Appendix G.2 for an explanation on their construction

³³With m^M representing the model moments and m^{Data} their respective data targets, simulated method of moments estimator is:

$$\hat{\psi} = \arg \min_o [m^M(o) - m^D]' W [m^M(o) - m^D] \quad (48)$$

where W is a positive definite weighting matrix. When RPS sampling variances are available, the baseline uses their inverse. Otherwise, the moments are weighted by the corresponding occupation and task employment shares.

follows:

$$q_{igt}^{AI} = \bar{q} \exp \left[\rho_1 (E_{ig} - \bar{E}) + \rho_2 (t - t_0) \right] \quad (49)$$

Where $\bar{q}$ is the baseline AI cost, $E_{ig} - \bar{E}$ is a zero centered measure of AI exposure of task i in occupation g . Here $\bar{q}$ governs the schedules level, ρ_1 its sensitivity to AI exposure and ρ_2 its speed of adjustment over time.

Model adoption probabilities. In order to map the model's discrete binary adoption decisions with the data reported adoption fractions at the task and occupation level I define the adoption probability of task i in occupation g to be $f_{igt} = \Pr(\Upsilon_{git} + \varepsilon_{git} \geq 0)$. Assuming that ε_{git} follows a logistic distribution with scale parameter $\tau > 0$. τ captures the unobserved differences in firms that affect their adoption decisions. Then f_{igt} is:

$$f_{igt} = \mathcal{L}(\Upsilon_{git}/\tau) \quad \text{with: } \mathcal{L}(x) = \frac{\exp(x)}{1 + \exp(x)} \quad (50)$$

Accordingly define the model task i adoption rate and the occupation g task adoption rate as

$$\beta_{it} = \sum_g \pi_{g|i} f_{igt}, \quad \beta_{gt} = \sum_i \pi_{i|g} f_{igt} \quad (51)$$

The mapping of β_{it} to its data counterpart is straightforward as the survey asks whether AI was used to perform task i . However, for β_{gt} the data counterpart measures whether a worker used AI for the job. To connect task adoption to this worker level indicator, define the probability that a worker in occupation g uses/adopts AI as:

$$u_{gt} = 1 - (1 - \beta_{gt})^\psi \quad (52)$$

where $\psi > 0$ governs the number of opportunities that a worker engages with an AI adopted task.

Identification and targeted moments. The aim is to estimate vector $o = \{\Theta, \bar{q}, \psi, \tau, \rho_1, \rho_2\}$. Identification is achieved by targeting a total of 306 moments coming from: $g = 1, \dots, 104$ average occupation adoption rates $\bar{u}_g$, plus $i = 1, \dots, 194$ average task adoption rates $\bar{\beta}_i$, plus 8 quarters of RPS broader group occupation adoption rates, namely: $u_{RPS} \Big|_{t=2026Q2}^{t=2024Q3}$.

Calibration of the rest of the parameters. The remaining aggregate parameters are externally calibrated or normalized. The baseline parameter choices are summarized in Table 3. The

preference and labor supply scales κ , ω , and A^Y are chosen to normalize 2024Q3 to be the initial reference point with $w_0 = p_0 = Y_0 = 1$.

Table 3: Calibrated parameters

Parameter	Description	Value	Source
σ	Task complementarity	0.50	From Acemoglu et al. (2025)
η	Composite good demand elasticity	0.70	From Comin et al. (2021)
ν	Labor supply elasticity	0.25	From Chetty et al. (2011)
r	Annual interest rate	4.0%	Standard calibration
κ	Labor supply level	2.43	Normalization so $w_0 = 1$
ω	Composite good demand level	1.00	Normalization so $Y_0 = 1$
A^Y	Composite good productivity	1.039	Normalization so $p_0 = 1$
A^M	Manual productivity	$1/\iota_g$	CPS permanent wage levels by g
p^{AI}	AI compute price	1.00	Normalization
f	Setup Labor (hours)	40.0	Assumed value (sensitivity see App F.1)
ε_B	Numeraire endowment scale	0.05	To ensure positive x_t .

Notes: The quarterly rate used is $r_q = (1.04)^{1/4} - 1 \simeq 0.00985$.

For this quantitative exercise, the relevant changes come through the AI variable unit cost, $q_{it}^{AI} = p_t^{AI} / A_{it}^{AI}$. I normalize $p_t^{AI} = 1$ for convenience and model changes in AI costs through A_{it}^{AI} . Equivalently, the same paths for q_{it}^{AI} can be interpreted as changes in the price of AI compute holding AI productivity fixed. These alternative decompositions generate identical equilibrium allocations and adoption decisions, with the only exception for the level of AI compute demand Z_t .

Because the numeraire endowment does not affect equilibrium allocations under quasilinearity, it is chosen nominally to ensure a positive numeraire consumption with $B = (1 + \varepsilon_B) \max_t \{p_t^{AI} Z_t\}$. Thus this sector mechanically follows the path of the AI sector $p_t^{AI} Z_t$.

Data Limitations. Inherited from the WORKBank dataset, the estimation sample covers occupations accounting for 15.25 percent of the U.S. wage/salaried employment. [Shao et al. \(2025\)](#) describe the sample as a broad and representative cross section at the sector level. However, the interpretation I adopt, is that of a selected sectoral sample. The results in Section 8 therefore characterize equilibrium responses only within this 15.25 percent covered segment.³⁴ Appendix E

³⁴Nationally representative time use data on the usage of AI in the workplace by task and occupation, would be useful to close this data gap. To the best of my knowledge such data set(s) are not publicly available. Upcoming tech enterprises such as Mercor, Micro1 among others have pioneered the collection of measurable AI use data in the work place.

conducts a national extrapolation, embedding a residual sector R , assumed not to be directly affected by AI, that accounts for the remaining 84.75 percent of U.S. wage/salaried employment. Adjusted equilibrium wages for this extrapolation are also included in Table 5, and fully reported in Appendix E.

8 Results

The estimated parameter values are $\hat{\Theta} = 20.724$, $\hat{q} = 46.896$, $\hat{\psi} = 2.449$, $\hat{\tau} = 722.565$, $\hat{\rho}_1 = -0.196$, $\hat{\rho}_2 = -0.064$.³⁵ Their interpretation is as follows:

- $\hat{\Theta} = 20.724$ means that an average occupation task pair $k = (i, g)$ requires about 20.724 hours of human validation per quarter after adopting AI. This is below the model implied validation threshold $\hat{\Theta}^*$, which exceeds 400 hours. Human validation is therefore too small to replace the productive labor displaced by AI. The simulated equilibrium consequently places the estimated economy on the wage decline side, case (43).
- $\hat{q} = 46.896$ means that producing one unit with AI costs about \$46.896. This is close to the respective average manual cost of about \$44.207. Thus, at the beginning of the sample 2024Q3, AI is approximately as expensive as manual production.
- A quarterly log decline of $\hat{\rho}_2 = -0.064$ translates into an annual AI-cost decline of 22.72% per year.
- A negative $\hat{\rho}_1$ means that activities classified as more exposed to AI are cheaper to automate.

Model fit. Table 4 shows the model fit towards the targeted moments. Because 306 moments are targeted, the table only reports the fit towards aggregate distributional moments of AI adoption by task and occupation, however it shows the precise targeted adoption rates by quarter for the RPS broader group occupations. Overall, the estimation delivers a satisfactory model fit.

³⁵A plot of the average estimated AI unit cost is shown in Figure 9.

Table 4: Model fit to targeted moments

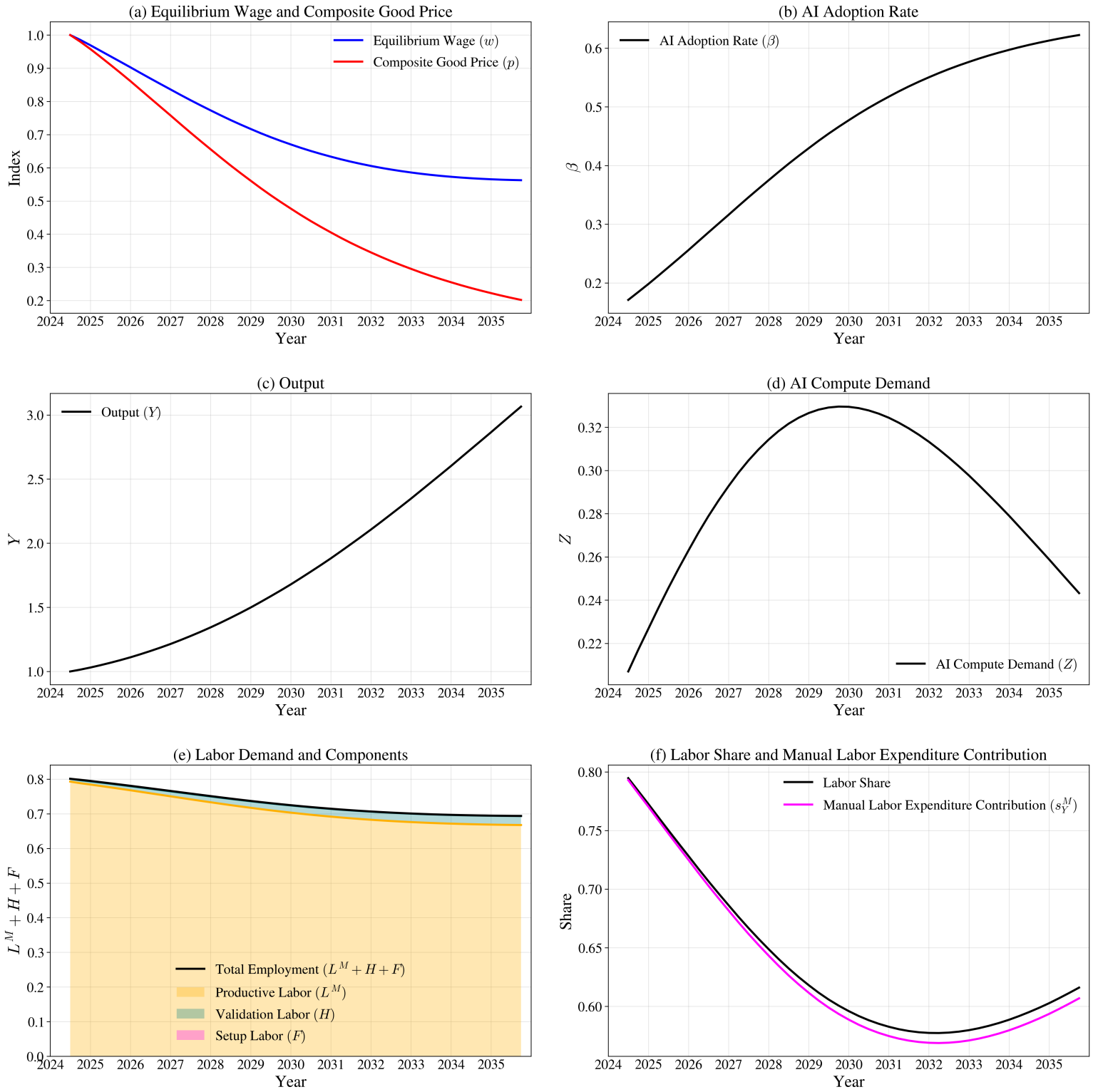
Block	Targeted moment	Data	Model	Difference
AI adoption rates by task ($\mathcal{I} = 104$)	Weighted Mean	0.263	0.272	0.009
	Weighted standard deviation	0.104	0.080	-0.024
	Weighted p10	0.138	0.152	0.015
	Weighted median	0.250	0.283	0.033
	Weighted p90	0.415	0.372	-0.043
AI adoption rates by occupation ($\mathcal{G} = 104$)	Weighted Mean	0.544	0.537	-0.006
	Weighted standard deviation	0.163	0.144	-0.019
	Weighted p10	0.303	0.337	0.035
	Weighted median	0.551	0.557	0.006
	Weighted p90	0.757	0.707	-0.050
AI adoption rates in RPS broader group occupations	$u_{RPS}(t = 2024Q3)$	0.404	0.359	-0.045
	$u_{RPS}(t = 2024Q4)$	0.382	0.391	0.009
	$u_{RPS}(t = 2025Q1)$	0.368	0.423	0.055
	$u_{RPS}(t = 2025Q2)$	0.438	0.456	0.018
	$u_{RPS}(t = 2025Q3)$	0.495	0.489	-0.007
	$u_{RPS}(t = 2025Q4)$	0.528	0.521	-0.007
	$u_{RPS}(t = 2026Q1)$	0.547	0.553	0.006
	$u_{RPS}(t = 2026Q2)$	0.563	0.584	0.021

Notes: Difference is model minus data. $\mathcal{I}$ and $\mathcal{G}$ are the total number of tasks and occupations.

Conditional forecast. Table 5 and Figure 6 report the model's conditional forecast under an assumed, 10 year extrapolated, further decline in AI costs. As AI becomes cheaper, the endogenous adoption share rises from 0.178 in 2024 to 0.618 in 2035. This substantially lowers the prices and increases output. Furthermore, the productivity gains due to AI adoption are labor saving. Productive labor usage declines from 0.791 to 0.668. Although validation labor increases, its increase is too small to offset manual labor displacement, as a result, total employment falls by approximately 13 percent and the equilibrium wage declines by 43 percent between 2024 and 2035.

The wage response is the result of the dominant labor displacement mechanism: each wave of adoption displaces more productive labor than it creates in validation and setup activities. The AI contribution to composite output rises through 2032, reaching approximately 43 percent, while the labor share falls. Thereafter, AI compute and its expenditure contribution decline despite continued adoption because falling AI unit costs allow a growing set of tasks to be produced with

Figure 6: Model forecasts: baseline sample



Notes: Forecast runs until 2035Q4.

less compute expenditure. The remaining manual tasks become relatively important bottlenecks, producing the modest recovery in both the manual contribution and the labor share after 2032.³⁶

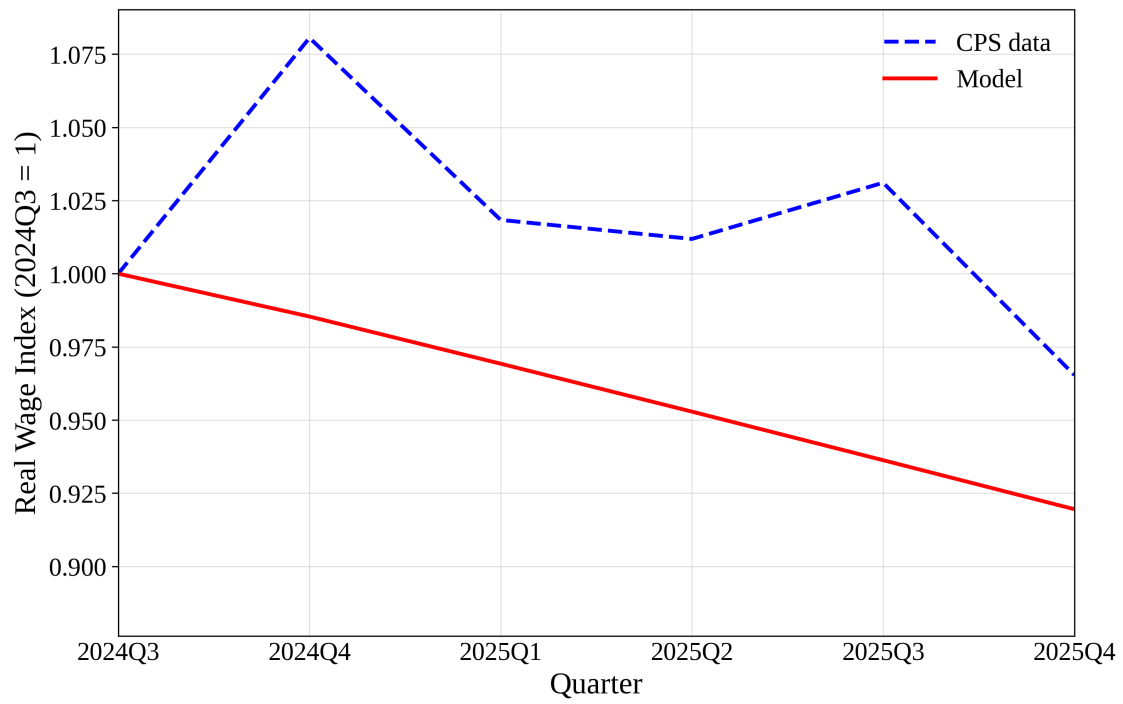
From Table 6, it can be seen that the national extrapolation results show mechanical attenuated effects: between 2024 and 2035, total employment falls by approximately 4.9 percent, the equilibrium wage declines by 18.2 percent, and output increases by 24 percent, compared with declines of 13 and 43 percent in employment and wages from the selected occupation sample. Nevertheless, the common wage falls because productive labor is displaced within the AI exposed sector. The displacement continues to exceed the validation and setup labor it creates. This exercise illustrates a scenario in which AI has more limited aggregate effects because a large share of economic activities either remains unexposed to AI or does not find AI sufficiently cost effective nor labor saving enough to adopt. As a consequence, even substantial adoption among exposed activities does not necessarily translate into large changes for the economy as a whole.

Untargeted evolution of real wages. Figure 7 shows that the model predicts a persistent wage decline, the observed CPS wage series though volatile exhibits a comparable downward sloping trend.³⁷ By 2025Q4, both series lie below their 2024Q3 levels.

³⁶These projections should not be interpreted as unconditional forecasts of the U.S. economy, rather as a conditional counterfactual based on an extrapolated AI cost path, holding other aggregate forces fixed.

³⁷Small sample volatility is a common known drawback of CPS data, however the CPS is the only available quarterly dataset that provided occupational wage and employment information to be linked to O*NET task characteristics.

Figure 7: Untargeted evolution of real wages



Notes: Expressed in 2025 U.S. dollars

Table 5: Model forecast

Years: 2024–2029

Variable	2024	2025	2026	2027	2028	2029
Endogenous AI adoption share β	0.1778	0.2196	0.2785	0.3383	0.3958	0.4480
Validation base $\overline{H}_A$	0.0004	0.0005	0.0006	0.0007	0.0009	0.0010
Equilibrium wage index w	0.9927	0.9445	0.8776	0.8121	0.7514	0.6987
Yearly wage change (%)	—	-4.8545	-7.0821	-7.4658	-7.4734	-7.0083
Productive labor L^M	0.7907	0.7783	0.7613	0.7440	0.7271	0.7118
Human validation labor $H = \Theta \overline{H}_A$	0.0084	0.0102	0.0128	0.0153	0.0176	0.0196
Operating labor demand $L^M + H$	0.7990	0.7885	0.7741	0.7593	0.7447	0.7315
Setup labor F	0.0006	0.0012	0.0012	0.0011	0.0010	0.0009
Total employment $L^M + H + F$	0.7996	0.7897	0.7753	0.7604	0.7458	0.7323
Labor share	0.7891	0.7559	0.7123	0.6718	0.6367	0.6091
Composite good price p	0.9899	0.9223	0.8225	0.7193	0.6196	0.5287
Composite good output Y	1.0072	1.0588	1.1474	1.2606	1.3997	1.5643
Manual contribution s_Y^M	0.7873	0.7532	0.7085	0.6670	0.6308	0.6023
AI contribution s_Y^{AI}	0.2128	0.2468	0.2915	0.3330	0.3692	0.3977
AI compute Z	0.2121	0.2408	0.2747	0.3015	0.3196	0.3283
Conditional threshold $\Theta^*(A)$	482.1472	469.1289	455.6219	446.1606	439.7343	435.1961
Nationwide Extrapolation						
Endogenous AI adoption share β	0.0273	0.0349	0.0459	0.0571	0.0672	0.0754
Equilibrium wage index w	0.9966	0.9732	0.9398	0.9071	0.8784	0.8561

Years: 2030–2035

Variable	2030	2031	2032	2033	2034	2035
Endogenous AI adoption share β	0.4929	0.5303	0.5607	0.5847	0.6034	0.6178
Validation base $\overline{H}_A$	0.0010	0.0011	0.0012	0.0012	0.0012	0.0013
Equilibrium wage index w	0.6559	0.6224	0.5976	0.5805	0.5697	0.5639
Yearly Wage change (%)	-6.1354	-5.0956	-3.9843	-2.8629	-1.8621	-1.0303
Productive labor L^M	0.6988	0.6882	0.6801	0.6742	0.6703	0.6681
Human validation labor $H = \Theta \overline{H}_A$	0.0213	0.0227	0.0238	0.0246	0.0253	0.0258
Operating labor demand $L^M + H$	0.7201	0.7109	0.7038	0.6988	0.6956	0.6939
Setup labor F	0.0007	0.0006	0.0005	0.0004	0.0003	0.0002
Total employment $L^M + H + F$	0.7209	0.7115	0.7043	0.6992	0.6959	0.6941
Labor share	0.5903	0.5800	0.5777	0.5826	0.5936	0.6093
Composite good price p	0.4493	0.3819	0.3258	0.2797	0.2423	0.2119
Composite good output Y	1.7529	1.9640	2.1951	2.4420	2.7002	2.9649
Manual contribution s_Y^M	0.5828	0.5719	0.5691	0.5737	0.5845	0.6001
AI contribution s_Y^{AI}	0.4172	0.4281	0.4309	0.4263	0.4155	0.3999
AI compute Z	0.3281	0.3206	0.3077	0.2908	0.2716	0.2511
Conditional threshold $\Theta^*(A)$	431.5676	428.2350	424.8029	421.0748	417.1068	412.9945
Nationwide Extrapolation						
Endogenous AI adoption share β	0.0819	0.0867	0.0903	0.0930	0.0950	0.0965
Equilibrium wage index w	0.8398	0.8288	0.8218	0.8178	0.8160	0.8157

Notes: All entries are annual averages of quarterly model simulations. The 2024Q3 equilibrium is normalized to $w = p = Y = 1$. For the 2026–2035 forecasts, occupational wage premia are held fixed, while the wage w_t always clears the labor market. The estimated AI-cost decline is extrapolated after 2026Q2.

9 Conclusion

This paper studies how AI adoption affects labor demand and wages when tasks operated with AI continue to require human involvement. I develop and estimate a general equilibrium task framework that explicitly models *human validation labor* as an input of the AI production technology. Human validation labor accounts for the work humans are still required to do even after full AI adoption: such as inspect, correct, authorize, or assume responsibility for the resulting output. The model identifies three key forces governing the labor market response to AI. (1) Adoption directly displaces productive labor from AI adopted tasks, (2) adopted tasks create recurring demand for human validation, (3) adoption lowers production costs and expands labor demand for the complementary tasks that remain manual. The paper shows that, when AI reduces productive labor in the aggregate, a validation threshold exists that determines the sign of the equilibrium response. Below this threshold, validation labor is insufficient to replace the productive hours displaced, causing employment and wages to fall; above the threshold, validation labor dominates displacement and both wages and employment increase. If the expansion in demand for complementary manual tasks is sufficiently large, AI can instead raise employment and wages even without validation labor. The main theoretical implication is therefore that even if economies adopt AI for the same tasks and obtain similar productivity gains, they can nevertheless experience opposite labor market outcomes.

To estimate the model and quantify these forces for the U.S. economy, I construct an occupation task data set that combines measures of human and AI involvement in the workplace, realized AI adoption, wage and employment, and task exposure to AI. The sample accounts for 15.25 percent of U.S. salaried employment, but broadly considered representative at the sector level. Observed adoption rates are used to infer the otherwise unobserved AI unit costs faced by firms and to estimate the level of human validation labor expressed in hours. The estimated validation requirement is approximately 20.7 hours per adopted occupation task pair per quarter, while the corresponding validation threshold remains above 400 hours. The U.S. sectors covered by the data sample are therefore placed on the labor displacement side of the threshold. Under a conditional ten year extrapolation of the estimated decline in AI unit costs, adoption among the occupations in the sample rises from 17.8 percent in 2024 to 61.8 percent in 2035. Output nearly triples, but employment falls by approximately 13 percent and the equilibrium wage declines by 43 percent. A nationwide extrapolation that incorporates the remaining 84.75 percent of the economy as a sector with no AI adoption naturally produces more attenuated effects: output rises by approximately 24 percent, employment falls by 4.9 percent, and the wage declines by 18.2 percent.

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A Proofs and Derivations

A.1 Proof of Lemma 1: Conditional Task Demand

Proof of Lemma 1. The Lagrangian for the conditional minimization problem 4.5 is (53). It follows, because all task costs are positive, so production constraint binds and it can be written as $\frac{1}{N} \sum_{i=1}^N y_i^\rho = Y^\rho$

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N q_i y_i + \lambda \left[\frac{1}{N} \sum_{i=1}^N y_i^\rho - Y^\rho \right] \quad (53)$$

Then, the first order condition for task i is

$$q_i + \lambda \rho y_i^{\rho-1} = 0 \quad (54)$$

Since $\rho - 1 = -1/\sigma$, define $b \equiv -\lambda \rho > 0$, then $q_i = b y_i^{-1/\sigma}$ or equivalently $y_i = b^\sigma q_i^{-\sigma}$.

To determine the level of task demand, substitute y_i into the binding production constraint:

$$Y^\rho = \frac{1}{N} \sum_{i=1}^N (b^\sigma q_i^{-\sigma})^\rho = b^{\sigma\rho} \underbrace{\frac{1}{N} \sum_{i=1}^N q_i^{1-\sigma}}_{=S} \quad \text{Define: } S \equiv \frac{1}{N} \sum_{i=1}^N q_i^{1-\sigma} \quad (55)$$

Solving for b^σ gives $b^\sigma = Y S^{\frac{\sigma}{1-\sigma}}$. Recall the variable unit cost of the task composite is (14)

$$m_{\mathcal{A}}(w) \equiv S^{\frac{1}{1-\sigma}} = \left[\frac{1}{N} \sum_{i=1}^N q_i(w; \mathcal{A})^{1-\sigma} \right]^{\frac{1}{1-\sigma}} \quad (56)$$

Then, substituting $S^{\frac{\sigma}{1-\sigma}} = m_{\mathcal{A}}(w)^\sigma$ and $b^\sigma = Y S^{\frac{\sigma}{1-\sigma}}$ into y_i delivers optimal task demand:

$$y_i = Y \left[\frac{m_{\mathcal{A}}(w)}{q_i(w; \mathcal{A})} \right]^\sigma \quad (57)$$

Finally, substituting optimal task demand into variable expenditure gives

$$\frac{1}{N} \sum_{i=1}^N q_i y_i = Y m_{\mathcal{A}}(w)^\sigma \left(\frac{1}{N} \sum_{i=1}^N q_i^{1-\sigma} \right) \quad (58)$$

$$= Y m_{\mathcal{A}}(w)^\sigma m_{\mathcal{A}}(w)^{1-\sigma} \quad (59)$$

$$= Y m_{\mathcal{A}}(w) \quad (60)$$

Hence $m_{\mathcal{A}}(w)$ is the minimum variable cost of producing one unit of the task composite. $\square$

A.2 Proof of Lemma 2: Decreasing Labor Demand

Proof of Lemma 2. By Shephard's lemma,

$$L^M(w, \mathcal{A}) = Y(w) \frac{\partial m_{\mathcal{A}}(w)}{\partial w} = \frac{Y(w) m_{\mathcal{A}}(w) s_M(w, \mathcal{A})}{w} \quad (61)$$

From (14),

$$\frac{\partial \log m_{\mathcal{A}}}{\partial \log w} = s_M \quad (62)$$

Moreover,

$$\frac{\partial \log s_M}{\partial \log w} = (1 - \sigma)(1 - s_M) \quad (63)$$

and (17) implies $\partial \log Y / \partial \log w = -\eta s_M$. Log differentiating (61) gives

$$\begin{aligned} \frac{\partial \log L^M}{\partial \log w} &= -\eta s_M + s_M + (1 - \sigma)(1 - s_M) - 1 \\ &= (\sigma - \eta) s_M - \sigma. \end{aligned}$$

If $\sigma \geq \eta$, the expression is no larger than $-\eta < 0$ because $s_M \leq 1$. If $\sigma < \eta$, it is smaller than $-\sigma < 0$. $\square$

A.3 Proof of Theorem 3: Labor Displacement and Scale Effect

Proof of Theorem 3. From (36)

1. If $\eta \leq \sigma$, then $\sigma - \eta \geq 0$. Since $\varrho_\beta \in (0, 1)$ then $(1 - \beta) \varrho_\beta^{\sigma - \eta} \leq 1 - \beta < 1$ hence productive labor demand falls.
2. If $\eta > \sigma$, then $(1 - \beta) \varrho_\beta^{(\sigma - \eta)} \gtrless 1$ if and only if $\varrho_\beta \gtrless (1 - \beta)^{1/(\eta - \sigma)}$

This establishes the three cases in (38)–(40). $\square$

A.4 Proof to Corollary 1: Aggregate Employment Response to AI adoption

Proof of Corollary 1. Labor market clearing implies

$$L^*(\mathcal{A}, \theta) = L(w(\mathcal{A}, \theta)). \quad (64)$$

Because labor supply is strictly increasing,

$$L'(w) = \frac{\nu}{w} L(w) > 0. \quad (65)$$

Therefore, equilibrium employment moves in the same direction as the equilibrium wage. $\square$

A.5 Proof of Theorem 4: The wage response

Proof of Theorem 4. Define excess labor demand after adoption at w_0 as:

$$\begin{aligned} G(w_0; \mathcal{A}, \theta) &= L_\beta^D(w_0) - L_0^D(w_0) \\ G(w_0; \mathcal{A}, \theta) &= \theta \bar{H}_\mathcal{A} - \underbrace{(L_0^D(w_0) - L_\beta^M(w_0))}_{\Delta_\beta^M} \\ G(w_0; \mathcal{A}, \theta) &= \theta \bar{H}_\mathcal{A} - \Delta_\beta^M \end{aligned} \quad (66)$$

From uniqueness 1, if $G(w_0; \mathcal{A}, \theta)$ is negative, labor demand at w_0 is below labor supply, so the wage must fall to clear the market. If it is positive, labor demand exceeds labor supply, so the wage must rise. If it is zero, w_0 continues to clear the labor market.

- If $\Delta_\beta^M > 0$. The two forces are equal at $G(w_0; \mathcal{A}, \theta) = 0$, namely when $\theta = \theta^*(\mathcal{A}) \equiv \frac{\Delta_\beta^M}{\bar{H}_\mathcal{A}}$. Therefore:
 - When $\theta < \theta^*(\mathcal{A})$: $G(w_0; \mathcal{A}, \theta) < 0$ then the wage falls after adoption.
 - When $\theta = \theta^*(\mathcal{A})$: $G(w_0; \mathcal{A}, \theta) = 0$ then the wage remains same.
 - When $\theta > \theta^*(\mathcal{A})$: $G(w_0; \mathcal{A}, \theta) > 0$ then the wage rises after adoption.
- If $\Delta_\beta^M = 0$. No productive labor is displaced, so adoption raises labor demand $G(w_0; \mathcal{A}, \theta) > 0$ at any $\theta > 0$ thus rising the equilibrium wage. At $\theta = 0$, $G(w_0; \mathcal{A}, \theta) = 0$, so wages remain unchanged.
- If $\Delta_\beta^M < 0$. Adoption itself increases productive labor demand and labor demand $G(w_0; \mathcal{A}, \theta) > 0$ thus rising wages at any $\theta \geq 0$.

$\square$

A.6 Decomposition of the composite output

From (6) partitioning tasks between the manual and AI methods gives

$$1 = \underbrace{\frac{1}{N} \sum_{i \notin \mathcal{A}} \left(\frac{y_i}{Y} \right)^\rho}_{s_Y^M} + \underbrace{\frac{1}{N} \sum_{i \in \mathcal{A}} \left(\frac{y_i}{Y} \right)^\rho}_{s_Y^{AI}}, \quad (67)$$

where $s_Y^M + s_Y^{AI} = 1$. Equivalently, using the conditional CES subaggregates,

$$s_Y^M = (1 - \beta) \left(\frac{Y^M}{Y} \right)^\rho, \quad s_Y^{AI} = \beta \left(\frac{Y^{AI}}{Y} \right)^\rho \quad (68)$$

By constant returns to scale, these terms are the corresponding Euler contributions:

$$s_Y^M = \frac{1}{Y} \sum_{i \notin \mathcal{A}} \frac{\partial Y}{\partial y_i} y_i, \quad s_Y^{AI} = \frac{1}{Y} \sum_{i \in \mathcal{A}} \frac{\partial Y}{\partial y_i} y_i \quad (69)$$

Under cost minimization, they also equal the manual and AI shares of variable production expenditure. Thus, (67) provides an exact additive decomposition of normalized composite output. Validation labor is not included because it is a fixed operating requirement, rather than a marginal input in the CES production aggregator.

B Computational Appendix

B.1 Computation of the stationary equilibrium and transitional dynamics

The stationary equilibrium and transitional dynamics from an initial condition are computed by forward iteration.

1. Initialize the economy from an initial adoption set $\mathcal{A}_{t=0}, \mathcal{N}_{t=0}$ and recover the equilibrium prices and quantities.
2. For date t , evaluate ξ_{it} and Υ_{it} for every task $i \notin \mathcal{A}_t$, using the inherited aggregate state.
3. Construct $\mathcal{N}_{t+1} = \{i \notin \mathcal{A}_t : \Upsilon_{it} \geq 0\}$ and update $\mathcal{A}_{t+1} = \mathcal{A}_t \cup \mathcal{N}_{t+1}$
4. Continue iterating until adoption converged, $\mathcal{N}_t = \emptyset$, $F_t = 0$, and $\max_{i \notin \mathcal{A}_t} \Upsilon_{it}(\mathcal{A}_t) < 0$. The resulting allocation is the terminal stationary equilibrium.

B.2 Pseudo calibration

The baseline parametrization for the numerical illustrations sets $N = 120$, $\sigma = 0.50$, $A^M = 1$, $\eta = 0.70$, $\omega = 1$, $\nu = 0.25$, $\kappa = 1$, $p^{AI} = 1$, $f_i = 0.50$, and $r = 0.04$. Validation difficulty h_i ranges from 0.75 to 1.25 across tasks and satisfies $N^{-1} \sum_i \bar{h}_i = 1$. The two economies differ only in validation intensity, with $\theta = 0.20$ and $\theta = 0.95$. For its transition numeraire endowment is set to $B = 0.0022$. All tasks initially have AI unit cost $q_i^{\text{pre}} = 5$. The terminal costs of the first 100 tasks are geometrically spaced from 3×10^{-7} to 3×10^{-6} ; the remaining 20 tasks range from 1 to 2. Task specific AI productivity is defined by $A_i^{AI} = p^{AI}/q_i^{\text{post}}$. The separate AI-price comparative static exercises in Figures 5 and 10 use $B = 0.045$ and hold AI productivity fixed while varying p^{AI} .

Along the transition, the cost of each task moves from its initial to its terminal level according to $q_{it}^{AI} = (q_i^{\text{pre}})^{1-g_{it}}(q_i^{\text{post}})^{g_{it}}$, where $g_{i0} = 0$ and $g_{it} = [1 + \exp\{-30(t - \tau_i)\}]^{-1}$ for $t \geq 1$. The first 100 tasks receive staggered τ_i values in 34 quarterly cohorts. In Figure 2, the order of these dates is reversed across the ready tasks, but the cost schedule is the same in both economies. Figure 8 displays the resulting decline in average AI unit cost. Adoption responds endogenously to these exogenous task-cost paths.

C Wealth effects of AI adoption

A simple extension replaces the linear utility from numeraire consumption with a strictly increasing and concave function $u(x)$:

$$U_t = u(x_t) + v(Y_t) - \chi(L_t), \quad u'(x) > 0, \quad u''(x) < 0 \quad (70)$$

The household first-order conditions become

$$v'(Y_t) = u'(x_t)p_t, \quad \chi'(L_t) = u'(x_t)w_t \quad (71)$$

Thus both composite-good demand and labor supply depend on household wealth, as reflected in numeraire consumption x_t . With the same labor disutility as in the baseline model,

$$L_t^s(w_t, x_t) = \left[\frac{u'(x_t)w_t}{\kappa} \right]^\nu \quad (72)$$

For example, if $u(x) = x^{1-\gamma}/(1-\gamma)$, then

$$L_t^s(w_t, x_t) = \left(\frac{w_t x_t^{-\gamma}}{\kappa} \right)^\nu, \quad \frac{\partial L_t^s}{\partial x_t} = -\frac{\gamma\nu}{x_t} L_t^s < 0 \quad (73)$$

An increase in wealth therefore reduces labor supplied at any given wage when leisure is a normal good.

Conditional on adoption set $\mathcal{A}$, equilibrium is determined by

$$L^s(w, x) = L^M(w, x, \mathcal{A}) + \theta \bar{H}_{\mathcal{A}} \quad (74)$$

Let $G(w, x) = L^s(w, x) - L^M(w, x, \mathcal{A}) - \theta \bar{H}_{\mathcal{A}}$. If composite-good demand is normal, then $\partial L^M / \partial x \geq 0$, and

$$\frac{\partial w}{\partial x} = -\frac{\partial L^s / \partial x - \partial L^M / \partial x}{\partial L^s / \partial w - \partial L^M / \partial w} > 0 \quad (75)$$

Greater wealth shifts labor supply inward and may also increase demand for the composite good; both channels raise the equilibrium wage. The response of equilibrium employment is no longer determined by the wage response alone, because wealth shifts both labor supply and labor demand.

If AI adoption raises wealth and leisure is a normal good, labor supply at the old wage falls:

$$L^s(w_0, x_1) < L^s(w_0, x_0)$$

If the composite good is also normal, greater wealth raises productive labor demand at the old wage. Both effects reduce $\theta_W^*(\mathcal{A}; x_1)$. Consequently, a smaller amount of validation labor is sufficient to prevent the wage from falling. Wealth effects therefore counteract the wage reducing effect of productive labor displacement.

D Additional Figures

Figure 8: Exogenous AI unit cost path

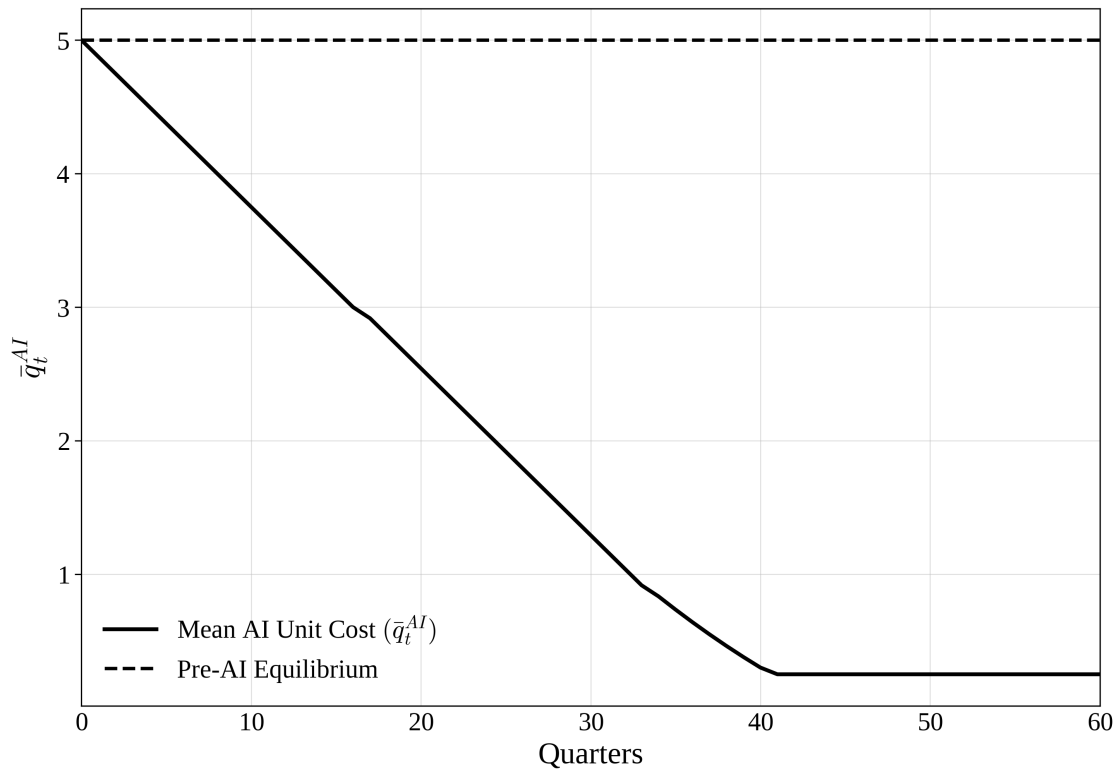


Figure 9: Estimated average AI unit cost

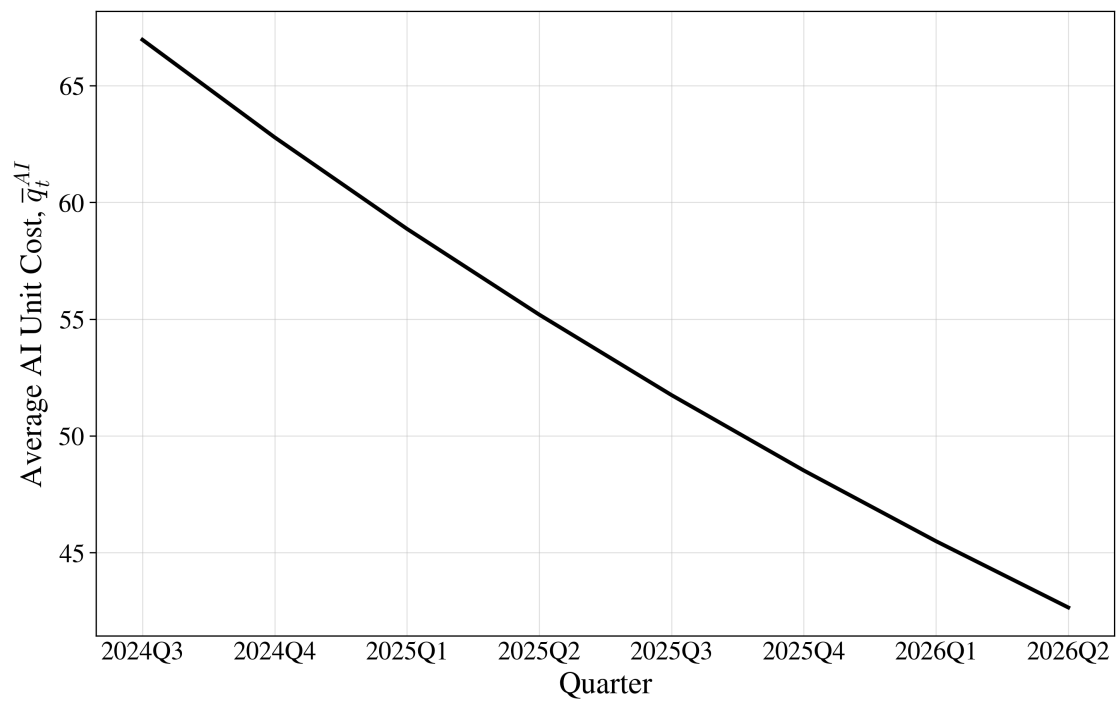
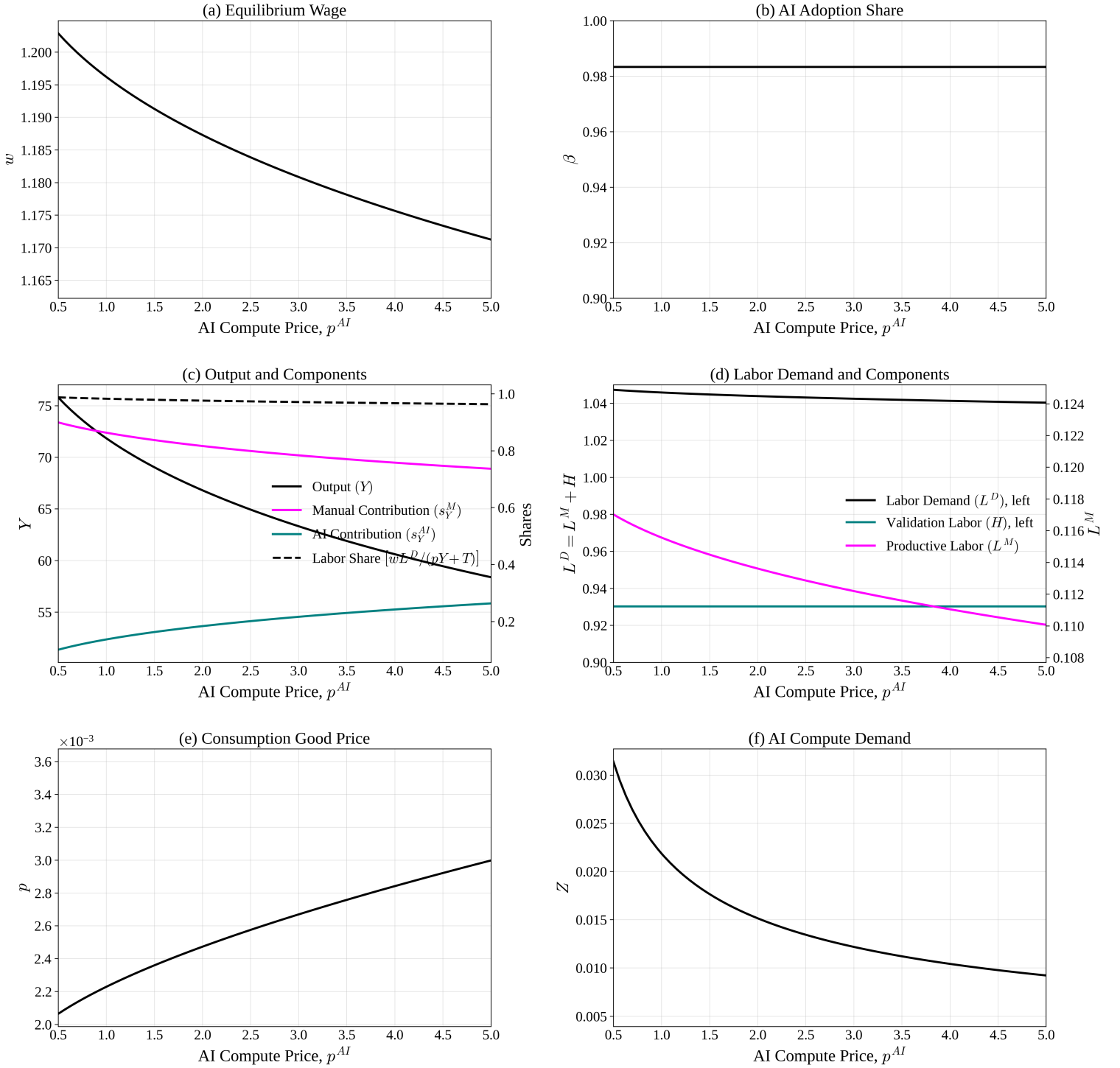


Figure 10: Increasing AI compute price, $\theta = 0.95$



Notes: The parametrization is $N = 120$, $\sigma = 0.50$, $A^M = 1$, $\eta = 0.70$, $\omega = 1$, $\nu = 0.25$, $\kappa = 1$, $\theta = 0.20$, $Nh_i = 1$, $f_i = 0.50$, $r = 0.04$, $q_0^{Pre-AI} = 5$, $B = 0.045$. $\bar{A}^{AI}$ is set to arbitrary high number. Simulations run for 80 periods, only steady state values are reported.

E National Extrapolation

This section extends the model by incorporating a residual sector R , that is held fixed and assumed not to be directly affected by AI. With this sector in place, the national labor market that the equilibrium wage solves is:

$$L(w_t) = (1 - \rho)L^D(w_t; \mathcal{A}_t) + \rho\bar{L}^R \quad \text{with: } \bar{L}^R = 1, \quad \frac{\partial L^R}{\partial w} = 0, \quad \rho = 0.8475$$

Model simulation results are presented in Table 6 and Figure 11 below:

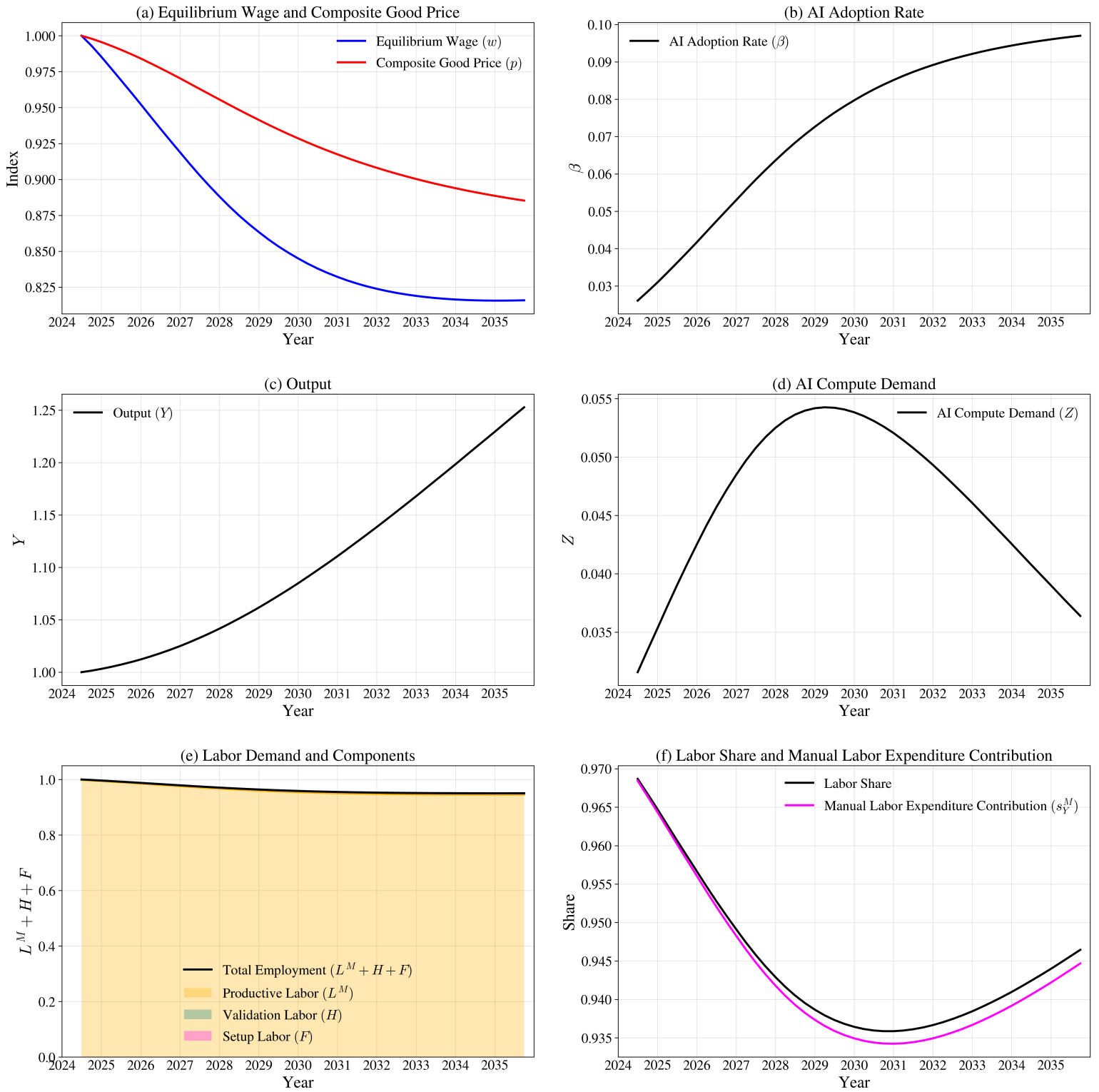
Table 6: National extrapolation of the model forecast

Years: 2024–2029						
Variable	2024	2025	2026	2027	2028	2029
Endogenous AI adoption share β	0.0273	0.0349	0.0459	0.0571	0.0672	0.0754
Validation base $\bar{H}_{\mathcal{A}}$	0.0001	0.0001	0.0001	0.0002	0.0002	0.0002
Equilibrium wage index w	0.9966	0.9732	0.9398	0.9071	0.8784	0.8561
Yearly wage change (%)	—	-2.3456	-3.4337	-3.4804	-3.1611	-2.5426
Productive labor L^M	0.9974	0.9909	0.9817	0.9725	0.9642	0.9577
Human validation labor $H = \Theta\bar{H}_{\mathcal{A}}$	0.0016	0.0020	0.0026	0.0032	0.0037	0.0041
Operating labor demand $L^M + H$	0.9990	0.9930	0.9843	0.9756	0.9679	0.9617
Setup labor F	0.0001	0.0003	0.0003	0.0003	0.0002	0.0002
Total employment $L^M + H + F$	0.9991	0.9932	0.9846	0.9759	0.9681	0.9619
Labor share	0.9677	0.9617	0.9538	0.9467	0.9411	0.9377
Composite good price p	0.9990	0.9916	0.9792	0.9649	0.9503	0.9365
Composite good output Y	1.0007	1.0062	1.0166	1.0308	1.0487	1.0700
Manual contribution s_Y^M	0.9674	0.9612	0.9531	0.9457	0.9400	0.9363
AI contribution s_Y^{AI}	0.0326	0.0388	0.0469	0.0543	0.0600	0.0637
AI compute Z	0.0325	0.0381	0.0449	0.0502	0.0533	0.0542
Conditional threshold $\Theta^*(\mathcal{A})$	482.1910	469.6790	457.5681	450.1301	445.5629	442.0255

Years: 2030–2035						
Variable	2030	2031	2032	2033	2034	2035
Endogenous AI adoption share β	0.0819	0.0867	0.0903	0.0930	0.0950	0.0965
Validation base $\bar{H}_{\mathcal{A}}$	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002
Equilibrium wage index w	0.8398	0.8288	0.8218	0.8178	0.8160	0.8157
Yearly wage change (%)	-1.8950	-1.3178	-0.8408	-0.4828	-0.2223	-0.0362
Productive labor L^M	0.9528	0.9495	0.9473	0.9461	0.9454	0.9453
Human validation labor $H = \Theta\bar{H}_{\mathcal{A}}$	0.0044	0.0046	0.0047	0.0049	0.0050	0.0050
Operating labor demand $L^M + H$	0.9572	0.9540	0.9520	0.9509	0.9504	0.9503
Setup labor F	0.0001	0.0001	0.0001	0.0001	0.0000	0.0000
Total employment $L^M + H + F$	0.9573	0.9541	0.9521	0.9510	0.9504	0.9504
Labor share	0.9361	0.9361	0.9373	0.9394	0.9421	0.9452
Composite good price p	0.9243	0.9138	0.9051	0.8978	0.8918	0.8869
Composite good output Y	1.0942	1.1209	1.1493	1.1792	1.2099	1.2410
Manual contribution s_Y^M	0.9346	0.9344	0.9355	0.9376	0.9403	0.9434
AI contribution s_Y^{AI}	0.0654	0.0656	0.0645	0.0624	0.0597	0.0566
AI compute Z	0.0533	0.0511	0.0481	0.0448	0.0412	0.0377
Conditional threshold $\Theta^*(\mathcal{A})$	438.5070	434.5550	430.1243	425.4122	420.5736	415.7192

Notes: All entries are annual averages of quarterly model simulations. The covered AI sector represents 15.25% of U.S. firm employment; the residual sector represents 84.75%, has zero AI adoption, and is held fixed at its 2024Q3 labor demand. Employment in each sector is normalized to one initially, and the common wage clears the national labor market. The initial national equilibrium is normalized to $w = p = Y = L = 1$. For 2026–2035, the estimated AI-cost decline is extrapolated after 2026Q2. Composite good prices, output, labor shares, and production contributions are employment share weighted national indexes. The conditional threshold refers to the sample covered AI sector.

Figure 11: Model forecasts: national extrapolation



Notes: Forecast runs until 2035Q4.

F Robustness

F.1 Changing the setup cost f

This section shows simulations varying the setup costs f . All results are robust. The results bellow confirm that f_{ig} would introduce relevant adoption heterogeneity across tasks, potentially capturing richer composition effects. I conduct two exercises, first a full re-estimation of the model structural parameters but setting the setup costs to $f = 320$ hours. Second, I simulate the baseline model but changing the setup from zero to 320 hours. The re-estimation yields $\hat{\Theta} = 14.651$, $\hat{q} = 43.578$, $\hat{\psi} = 2.471$, $\hat{\tau} = 747.640$, $\hat{\rho}_1 = -0.154$, $\hat{\rho}_2 = -0.068$, and Table 7 shows the conditional forecast results. Table 8 shows that changing the setup cost without re-estimating the model mechanically reduces adoption and thus the decline in wages and employment.

Table 7: Model forecast using the re-estimated $f = 320$ set up cost

Years: 2024–2029						
Variable	2024	2025	2026	2027	2028	2029
Endogenous AI adoption share β	0.1773	0.2208	0.2805	0.3401	0.3958	0.4447
Validation base $\bar{H}_A$	0.0004	0.0005	0.0006	0.0007	0.0009	0.0009
Equilibrium wage index w	0.9973	0.9513	0.8836	0.8182	0.7592	0.7102
Yearly wage change (%)	–	-4.6177	-7.1103	-7.4114	-7.2053	-6.4551
Productive labor L^M	0.7932	0.7785	0.7620	0.7459	0.7313	0.7188
Human validation labor $H = \Theta \bar{H}_A$	0.0059	0.0073	0.0092	0.0109	0.0125	0.0138
Operating labor demand $L^M + H$	0.7992	0.7858	0.7712	0.7569	0.7438	0.7326
Setup labor F	0.0054	0.0093	0.0094	0.0088	0.0077	0.0065
Total employment $L^M + H + F$	0.8046	0.7951	0.7806	0.7657	0.7515	0.7391
Labor share	0.7951	0.7624	0.7203	0.6824	0.6512	0.6292
Composite good price p	0.9933	0.9233	0.8172	0.7090	0.6069	0.5160
Composite good output Y	1.0047	1.0580	1.1527	1.2735	1.4202	1.5910
Manual contribution s_Y^M	0.7927	0.7585	0.7154	0.6767	0.6450	0.6227
AI contribution s_Y^{AI}	0.2073	0.2415	0.2846	0.3233	0.3550	0.3773
AI compute Z	0.2068	0.2356	0.2677	0.2915	0.3054	0.3093
Conditional threshold $\Theta^*(A)$	474.8708	462.0490	449.0309	440.1021	433.9570	429.3369

Years: 2030–2035						
Variable	2030	2031	2032	2033	2034	2035
Endogenous AI adoption share β	0.4857	0.5192	0.5456	0.5659	0.5815	0.5933
Validation base $\bar{H}_A$	0.0010	0.0011	0.0011	0.0012	0.0012	0.0012
Equilibrium wage index w	0.6717	0.6430	0.6231	0.6104	0.6033	0.6006
Yearly wage change (%)	-5.4144	-4.2737	-3.1018	-2.0380	-1.1546	-0.4627
Productive labor L^M	0.7087	0.7010	0.6956	0.6922	0.6903	0.6896
Human validation labor $H = \Theta \bar{H}_A$	0.0149	0.0157	0.0164	0.0169	0.0173	0.0176
Operating labor demand $L^M + H$	0.7236	0.7168	0.7120	0.7091	0.7076	0.7073
Setup labor F	0.0053	0.0042	0.0033	0.0025	0.0019	0.0015
Total employment $L^M + H + F$	0.7289	0.7210	0.7153	0.7116	0.7096	0.7087
Labor share	0.6165	0.6125	0.6165	0.6271	0.6429	0.6625
Composite good price p	0.4384	0.3736	0.3206	0.2777	0.2433	0.2156
Composite good output Y	1.7834	1.9944	2.2196	2.4538	2.6917	2.9286
Manual contribution s_Y^M	0.6098	0.6058	0.6099	0.6206	0.6366	0.6564
AI contribution s_Y^{AI}	0.3902	0.3942	0.3901	0.3794	0.3634	0.3436
AI compute Z	0.3046	0.2933	0.2773	0.2583	0.2378	0.2169
Conditional threshold $\Theta^*(A)$	425.3557	421.4396	417.2597	412.8314	408.2595	403.6433

Notes: Model re-estimation and conditional forecast with setup cost $f = 320$. All entries are annual averages of quarterly model simulations.

Table 8: Setup cost robustness: $f = [0, 20, 40, 80, 160, 320]$

Variable	$f = 0$			$f = 20$			$f = 40$		
	2024	2029	2035	2024	2029	2035	2024	2029	2035
Endogenous AI adoption share β	0.19	0.46	0.63	0.18	0.45	0.62	0.18	0.45	0.62
Validation base $\overline{H}_A$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Equilibrium wage index w	0.99	0.69	0.56	0.99	0.70	0.56	0.99	0.70	0.56
Productive labor L^M	0.78	0.70	0.66	0.79	0.71	0.66	0.79	0.71	0.67
Human validation labor $H = \Theta \overline{H}_A$	0.01	0.02	0.03	0.01	0.02	0.03	0.01	0.02	0.03
Operating labor demand $L^M + H$	0.79	0.72	0.68	0.80	0.73	0.69	0.80	0.73	0.69
Setup labor F	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Total employment $L^M + H + F$	0.79	0.72	0.68	0.80	0.73	0.69	0.80	0.73	0.69
Labor share	0.78	0.60	0.60	0.78	0.60	0.60	0.79	0.61	0.61
Composite-good price p	0.99	0.52	0.21	0.99	0.53	0.21	0.99	0.53	0.21
Composite output Y	1.01	1.58	3.03	1.01	1.57	3.00	1.01	1.56	2.96
Manual contribution s_Y^M	0.78	0.59	0.59	0.78	0.60	0.59	0.79	0.60	0.60
AI contribution s_Y^{AI}	0.22	0.41	0.41	0.22	0.40	0.41	0.21	0.40	0.40
AI compute Z	0.22	0.34	0.26	0.22	0.33	0.25	0.21	0.33	0.25
Conditional threshold $\Theta^*(\mathcal{A})$	481.6	435.5	414.0	481.9	435.4	413.5	482.1	435.2	413.0

Variable	$f = 80$			$f = 160$			$f = 320$		
	2024	2029	2035	2024	2029	2035	2024	2029	2035
Endogenous AI adoption share β	0.17	0.44	0.61	0.16	0.42	0.59	0.13	0.38	0.54
Validation base $\overline{H}_A$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Equilibrium wage index w	0.99	0.71	0.57	1.00	0.72	0.59	1.00	0.75	0.62
Productive labor L^M	0.80	0.72	0.68	0.81	0.74	0.70	0.84	0.77	0.73
Human validation labor $H = \Theta \overline{H}_A$	0.01	0.02	0.03	0.01	0.02	0.02	0.01	0.02	0.02
Operating labor demand $L^M + H$	0.81	0.74	0.70	0.82	0.76	0.72	0.85	0.79	0.76
Setup labor F	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.01	0.00
Total employment $L^M + H + F$	0.81	0.74	0.70	0.82	0.76	0.72	0.85	0.79	0.76
Labor share	0.80	0.62	0.62	0.81	0.64	0.64	0.84	0.68	0.69
Composite-good price p	0.99	0.54	0.22	0.99	0.55	0.23	1.00	0.58	0.26
Composite output Y	1.01	1.55	2.90	1.01	1.52	2.78	1.00	1.46	2.54
Manual contribution s_Y^M	0.80	0.61	0.61	0.81	0.63	0.64	0.84	0.68	0.68
AI contribution s_Y^{AI}	0.20	0.39	0.39	0.19	0.37	0.36	0.16	0.32	0.32
AI compute Z	0.20	0.32	0.25	0.19	0.31	0.23	0.16	0.27	0.21
Conditional threshold $\Theta^*(\mathcal{A})$	482.7	434.8	412.0	483.7	434.1	409.9	485.5	432.5	405.7

Notes: Entries are annual averages of quarterly model simulations. The model is not re-estimated.

G Data Appendix

This section describes the steps taken to process and construct the baseline occupation task data set for the U.S.

G.1 Source files and samples

Table 9 summarizes the source files and their uses. WORKBank provides worker and expert ratings used to proxy human validation requirements. O*NET supplies the task hierarchy and baseline activity weights. RPS and the associated FRED series provide activity, occupation, and quarterly AI-adoption outcomes. CPS provides wages, employment, and usual working hours; OEWS employment data helps allocate CPS occupations to the chosen detailed O*NET hierarchy. [Eloundou et al. \(2024\)](#) provide task exposure labels.

Table 9: Source files and variables used

Source	Sample used	Variables used	Use
WORKBank	January-May 2025; 5,709 worker and 2,048 expert ratings.	O*NET occupation and Task IDs. Ratings: human agency, uncertainty, domain expertise, automation capacity.	Baseline human validation measures for 104 occupations and 158 IWAs.
O*NET 31.0	August 2026; full occupation-task database.	Task frequency, DWA-IWA-GWA hierarchy.	IWA task nomenclature and task weights.
RPS	Four pooled quarterly waves: August 2025 to May 2026. Quarterly series 2024Q3–2026Q2.	IWA activity adoption; occupation worker use; major group and national quarterly work use.	Calculation of the 194 activity, 104 occupations, and eight quarterly AI adoption targets.
IPUMS CPS	2024-2025 Basic Monthly: 23 available months. Rotation group eligible workers.	Occupation; employment; earning weights; usual working hours; weekly earnings;	Employment weights, weekly hours, wages.
BLS and FRED	May 2024 OEWS; SOC-CPS crosswalk; monthly 2024-2025 CPI.	SOC employment and wages; occupation mappings; Inflation.	CPS to SOC occupation merge, and conversion to 2025 U.S. dollars.
Eloundou et al.	Full O*NET task label set.	Human and GPT-4 exposure classification.	Calculation of occupation-IWA exposure measures.

Notes: October 2025 missing due to government shut down.

G.2 WORKBank-O*NET matching and activity weights

The matching procedure consists of pairing WORKbank tasks to O*NET 31.0 task nomenclature using their O*NET-SOC code and O*NET Task ID. Of the 2,131 tasks in the broader WORKBank metadata, 2,118 match O*NET 31.0 exactly. These amount to 22 worker responses excluded. Among the 846 tasks in the worker expert audited task database, 842 match exactly. The four unmatched audited tasks are excluded and correspond to retired or changed tasks between O*NET vintages, results in 9 expert ratings excluded. The matching procedure left 5,709 worker responses and 2,048 expert ratings.

The goal is to distinguish validation requirements associated with an occupation from those associated with the tasks performed by its workers. WORKBank rates specific tasks within each occupation. To compare those ratings across occupations, I group the tasks using O*NET's Intermediate Work Activities (IWAs). These broader activity categories link all 104 occupations³⁸, allowing us to estimate occupation and activity differences on a common scale. Using O*NET's narrower Detailed Work Activities does not link the full sample of occupations, thus the use of IWA's maintains reasonable task level heterogeneity while maximizing the respective occupational matching.

Allocating work hours across activities. AI adoption decisions depend partly on how many manual work hours an activity uses: automating an activity that uses many hours can save more labor than automating one that uses few. CPS reports usual weekly hours by occupation, but does not divide those hours among the occupation's activities. The weight $\pi_{i|g}$ makes this allocation. If occupation g has $\bar{u}_g$ usual weekly hours, the baseline weekly manual hours proxy for activity i is $\ell_{ig}^M = \bar{u}_g \pi_{i|g}$. Because activity level hours are not observed, $\pi_{i|g}$ is constructed from O*NET relevance and frequency ratings as follows: let each task j in occupation g receive a score $a_{jg} = R_{jg} F_{jg}$ where R_{jg} measures how relevant the task is to the occupation and F_{jg} is its average frequency category score (how often a task is performed). A task linked to several Detailed Work Activities (DWAs) has its score divided equally among those links. The scores are then added within each Intermediate Work Activity (IWA).

Let S_{ig} be the total score assigned to IWA i in occupation g . Its share of the occupation's allocated work hours is

$$\pi_{i|g} = \frac{S_{ig}}{\sum_{k \in \mathcal{I}_g} S_{kg}}, \quad \sum_{i \in \mathcal{I}_g} \pi_{i|g} = 1 \quad (76)$$

An activity assigned a larger score thus receives a larger share of the occupation's baseline working hours. This is a proxy, not an observed time share or a measure of AI validation hours. Missing relevance or frequency ratings are replaced by the occupation median, or by the full sample median when the occupation median is unavailable. For future work, the ideal replacement of these intensity proxies are directly observed occupation task time use measures.

G.3 WORKBank Relative human validation profiles

The theory requires a measure for human validation requirements by occupation and activity pair. WORKBank does not measure validation hours, but its ratings can be used to proxy the activities that require more human involvement than others. The baseline proxy combines three task ratings from expert responses.³⁹ That is the responses of expert n on activity i in occupation g :

$$v_{nig} = \left(V_{nig}^{\text{agency}} V_{nig}^{\text{uncertainty}} V_{nig}^{\text{expertise}} \right)^{1/3}. \quad (77)$$

³⁸The 104 occupations cover management; business and financial operations; computer, mathematical, and information technology work; architecture and engineering; life, physical, and social sciences; legal services; education and libraries; arts, design, writing, and media; healthcare; sales; and office and administrative support. The sample is therefore concentrated in knowledge intensive, professional, technical, and administrative occupations.

³⁹A rated task can belong to more than one IWA. In that case, its response weight is divided equally across those IWAs, so the same rating is not counted multiple times.

The ratings describe the desired human agency, uncertainty, and required domain expertise; each ranges from one to five. A higher value indicates a greater need for human validation. Worker level human agency rating provides an alternative proxy left for robustness.

G.4 RPS survey adoption rates

The RPS activity adoption rate is the survey weighted share of performed activities for which a respondent reports using Generative AI. A respondent who reports no job related GenAI use contributes zero to all performed activities. All 322 IWA titles in the supplied RPS file match a unique current O*NET IWA title. The RPS Gen AI Adoption Tracker files provide 256 non-suppressed IWA adoption rates.⁴⁰ Of these, 194 correspond to IWAs in the 104 occupation sample.

The occupation target is the survey weighted fraction of workers who answer that they use generative AI for their job. For each O*NET occupation, the adoption target is taken from the smallest RPS occupation group with a reported worker use rate. The broad occupation rate is used when available; otherwise the minor group rate is used, followed by the major group rate. This yields one target for each of the 104 model occupations. These worker use rates are not treated as task adoption rates; the parameter ψ that maps these two is estimated in Section 7.

The quarterly RPS/FRED file reports work use by major occupation group. Let u_{mt}^{RPS} denote the reported rate for major group m and let $\pi_m^S = \sum_{g \in m} \pi_g$ be that group's employment share among the 104 selected occupations. The selected sample time series target is then

$$u_t^S = \sum_m \pi_m^S u_{mt}^{RPS}, \quad t = 2024Q3, \dots, 2026Q2 \quad (78)$$

G.5 CPS labor market statistics and occupation matching

CPS microdata is used to compute the required labor market statistics. Employment uses all employed respondents and the respective monthly weight. Firm employment restricts the sample to salaried workers. Occupation hours are the weighted mean of usual weekly hours among all employed workers. Wages use earnings eligible respondents in outgoing rotation groups and the CPS earnings weight. For non hourly workers, hourly earnings are weekly earnings divided by usual weekly hours at the main job. I trim the weighted national wage distribution at its first and ninety ninth percentiles and deflate nominal wages by monthly CPI. The base is the mean index across the eleven observed CPS months in 2025, $\overline{CPI}_{2025} = 321.943$.

Assigning CPS statistics to O*NET occupations. CPS uses a different occupation classification compared to the O*NET nomenclature used for the model. A CPS occupation may contain several SOC occupations. The BLS crosswalk identifies which SOC occupations belong to each CPS occupation. When CPS occupation c is linked to several SOC occupations, I use OEWS May 2024 employment shares to determine how its employment is divided:

$$\lambda_{s|c} = \frac{E_s^{OEWS}}{\sum_{s' \in \mathcal{S}(c)} E_{s'}^{OEWS}}, \quad s \in \mathcal{S}(c) \quad (79)$$

⁴⁰RPS suppresses rates based on fewer than 20 survey observations.

Where $\mathcal{S}(c)$ is the set of SOC occupations linked to CPS occupation c . Thus a SOC occupation representing a larger share of OEWS employment receives a larger share of the CPS total.

Some SOC occupations are further divided into several O*NET occupations, which the available employment data cannot distinguish. Thus, their assigned CPS employment is divided equally among all current O*NET occupations under that SOC code, including those outside the 104 occupation sample. Then for occupation g under SOC code $s(g)$, the resulting employment is:

$$E_g = \frac{1}{n_{s(g)}} \sum_{s(g) \in \mathcal{S}(c)} E_c^{CPS} \lambda_{s(g)|c} \quad (80)$$

where E_c^{CPS} is average monthly CPS salaried employment and $n_{s(g)}$ is the number of current O*NET occupations under that SOC code.

Finally, the sample's occupation and occupation task weights are:

$$\pi_g = \frac{E_g}{\sum_{g' \in \mathcal{G}} E_{g'}}, \quad \pi_{ig} = \pi_g \pi_{i|g}, \quad \sum_g \pi_g = \sum_g \sum_{i \in \mathcal{I}_g} \pi_{ig} = 1 \quad (81)$$

From this matching we learn that the WORKBank 104 occupations represent 15.253 percent of average 2024–2025 U.S. salaried employment.

G.6 Task exposure

[Eloundou et al. \(2024\)](#) task labels are joined to O*NET 31.0 data, by O*NET-SOC code and Task ID, mapped from task statements to DWAs and then to IWAs, and then averaged within each occupation IWA pair. The baseline assigns scores of zero, one, and one half to $E0$, $E1$, and $E2$ and averages them within the task occupation pairs. A missing pair is imputed by the respective IWA mean across occupations and then, if necessary, the overall mean. Human labels are used as the baseline. GPT-4 labels are kept as robustness measure.